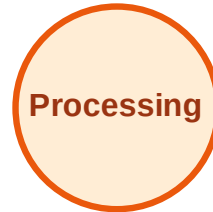


BFS Traversal

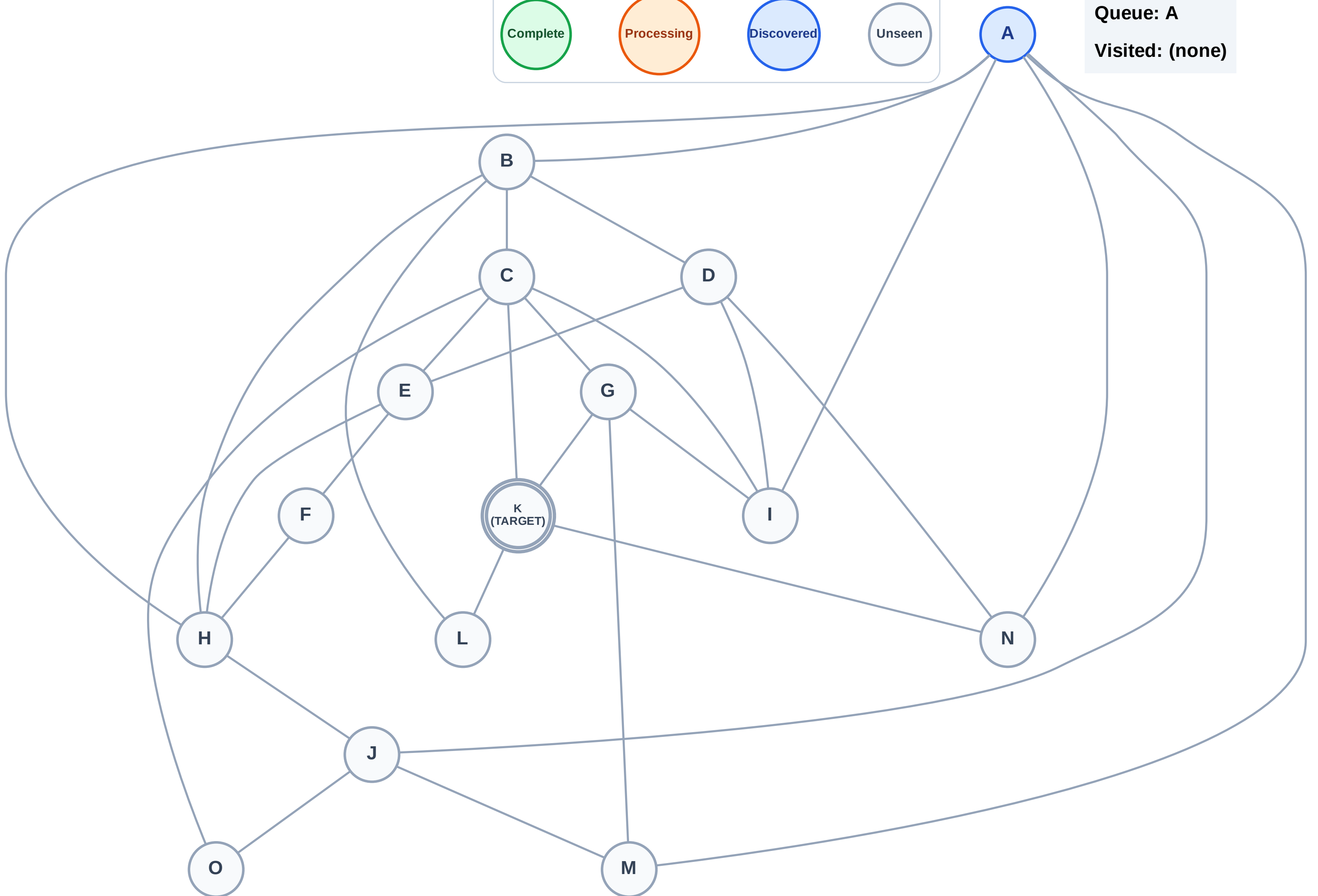
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

Visited: (none)



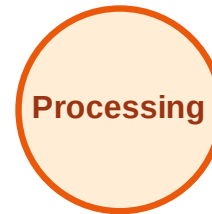
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



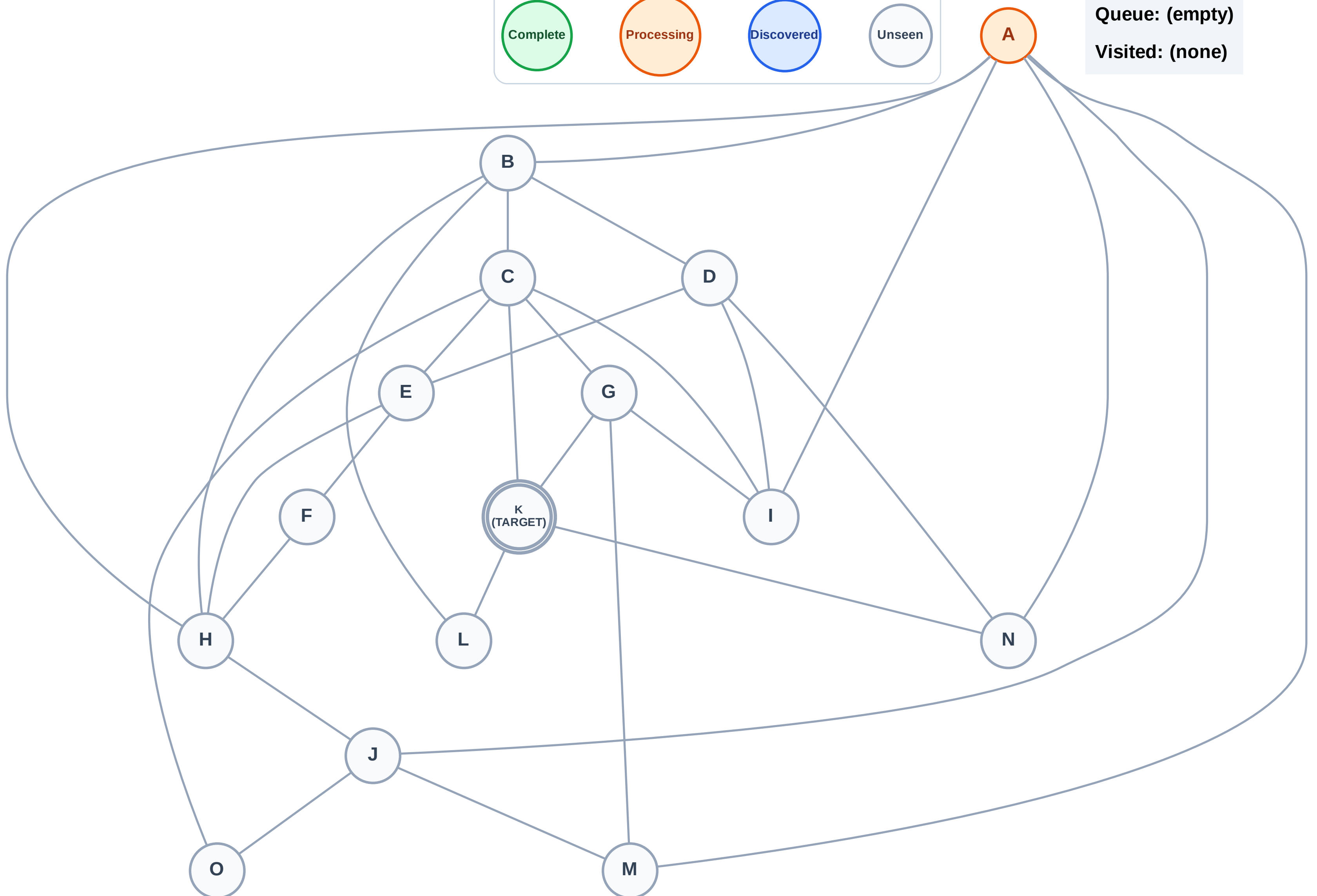
Discovered



Unseen

Queue: (empty)

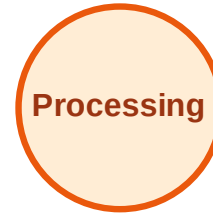
Visited: (none)



BFS Traversal

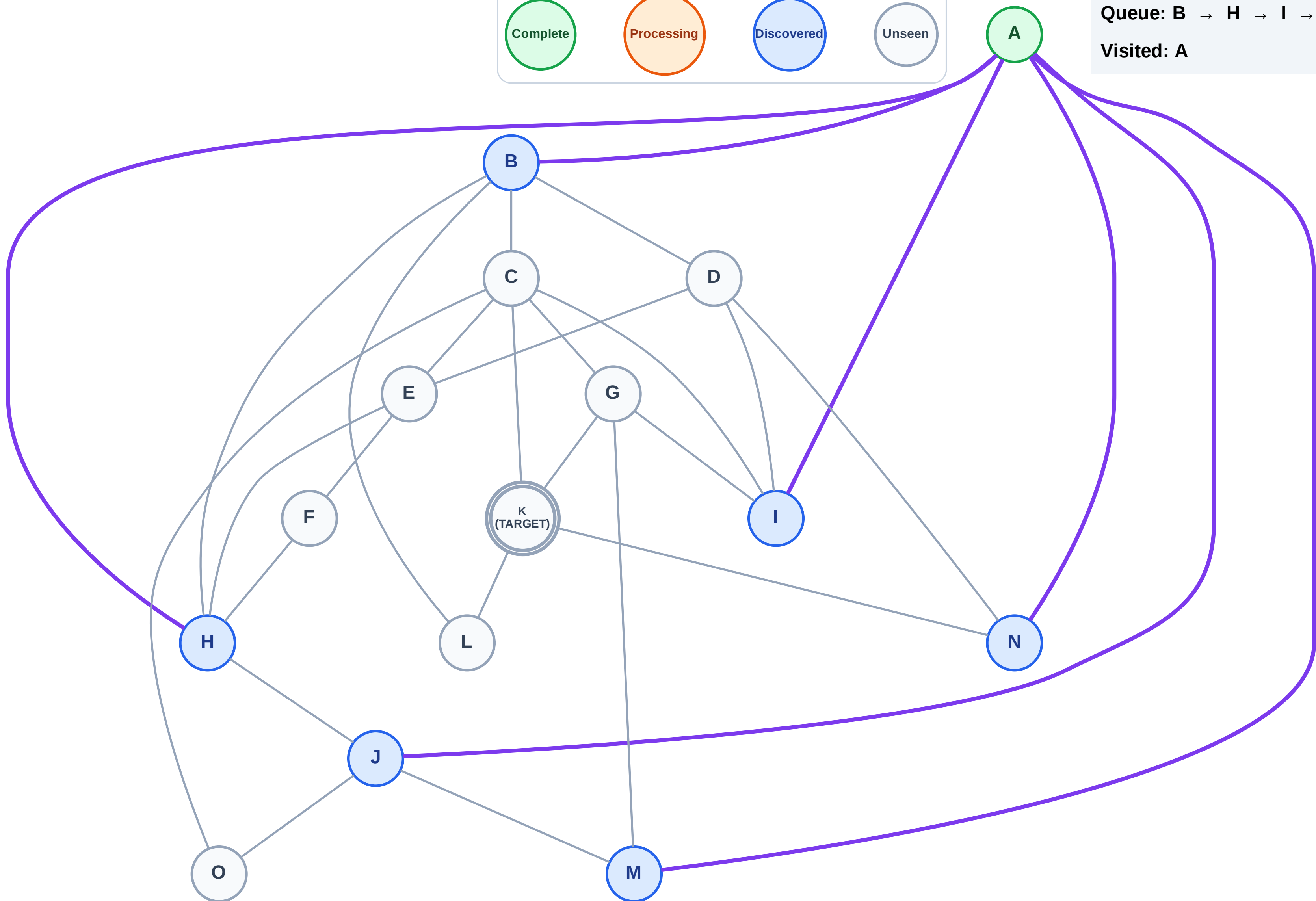
Step 3: Discover B, H, I, J, M, N from A

Legend



Queue: B → H → I → J → M → N

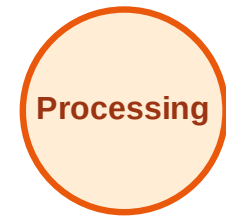
Visited: A



BFS Traversal

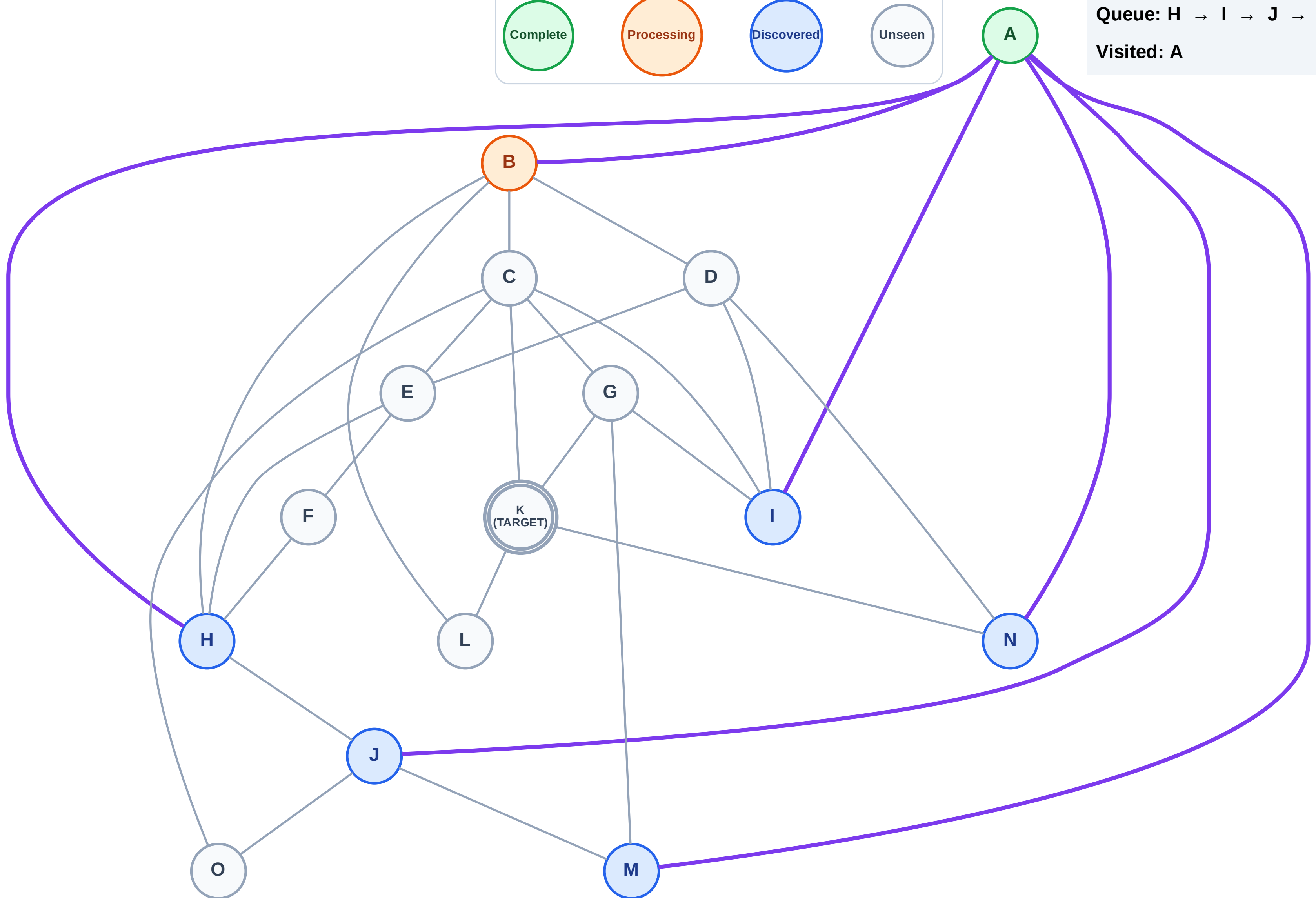
Step 4: Process node B

Legend



Queue: H → I → J → M → N

Visited: A



BFS Traversal

Step 5: Discover C, D, L from B

Legend



Complete



Processing

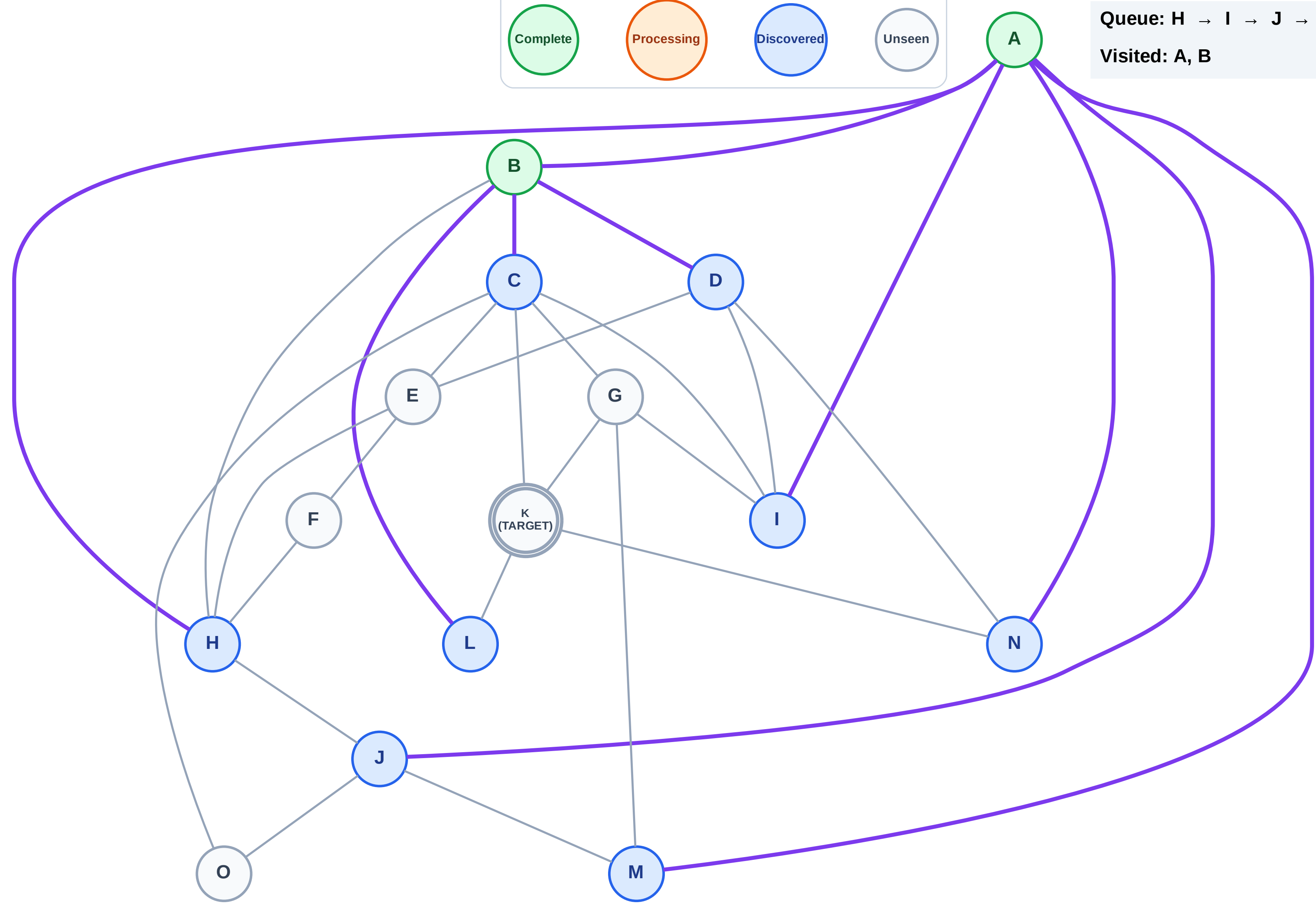


Discovered



Unseen

Queue: H → I → J → M → N → C → D → L
Visited: A, B



BFS Traversal

Step 6: Process node H

Legend

Complete

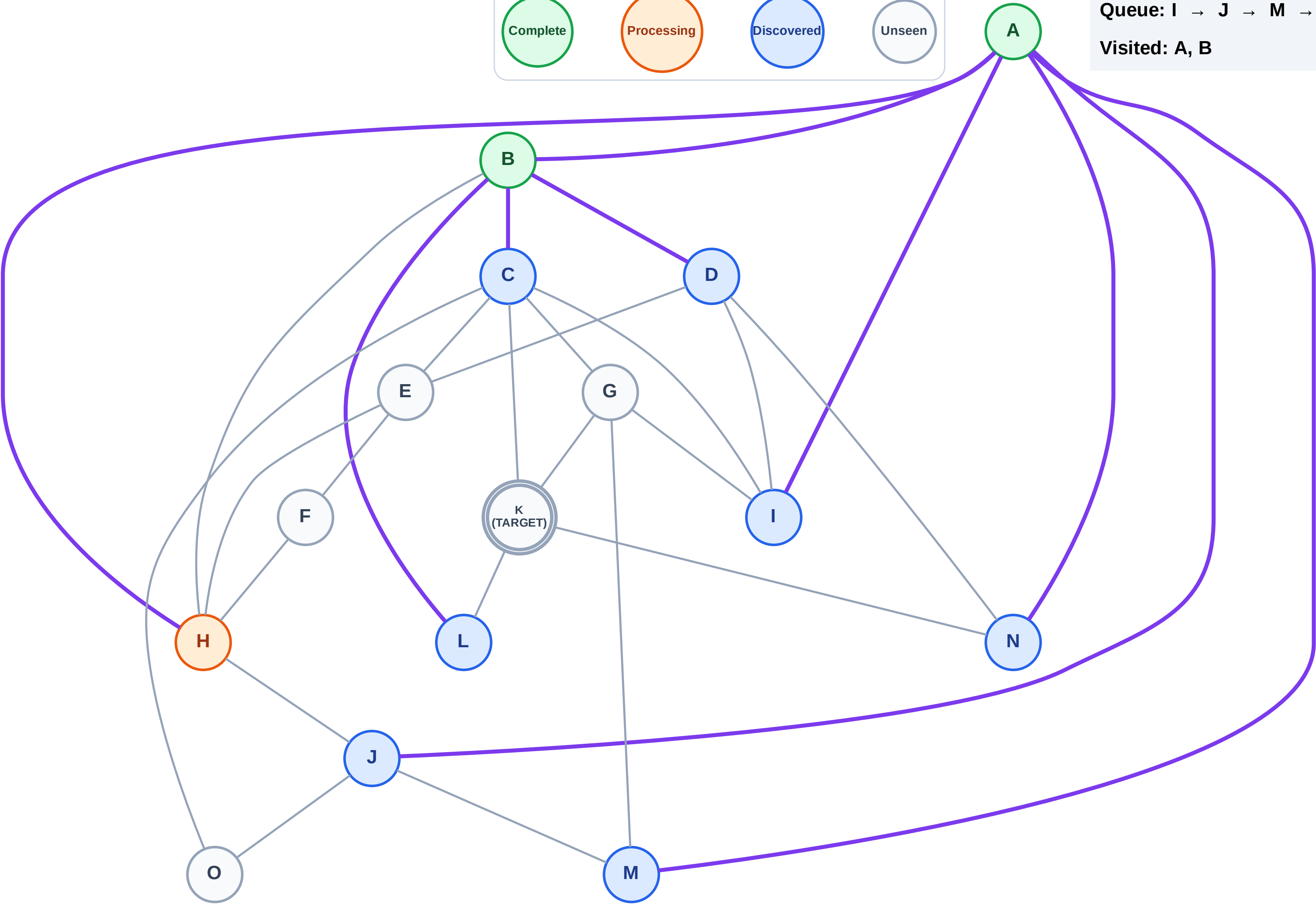
Processing

Discovered

Unseen

Queue: I → J → M → N → C → D → L

Visited: A, B



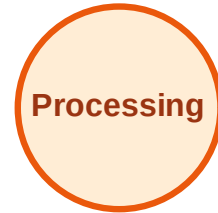
BFS Traversal

Step 7: Discover E, F from H

Legend



Complete



Processing



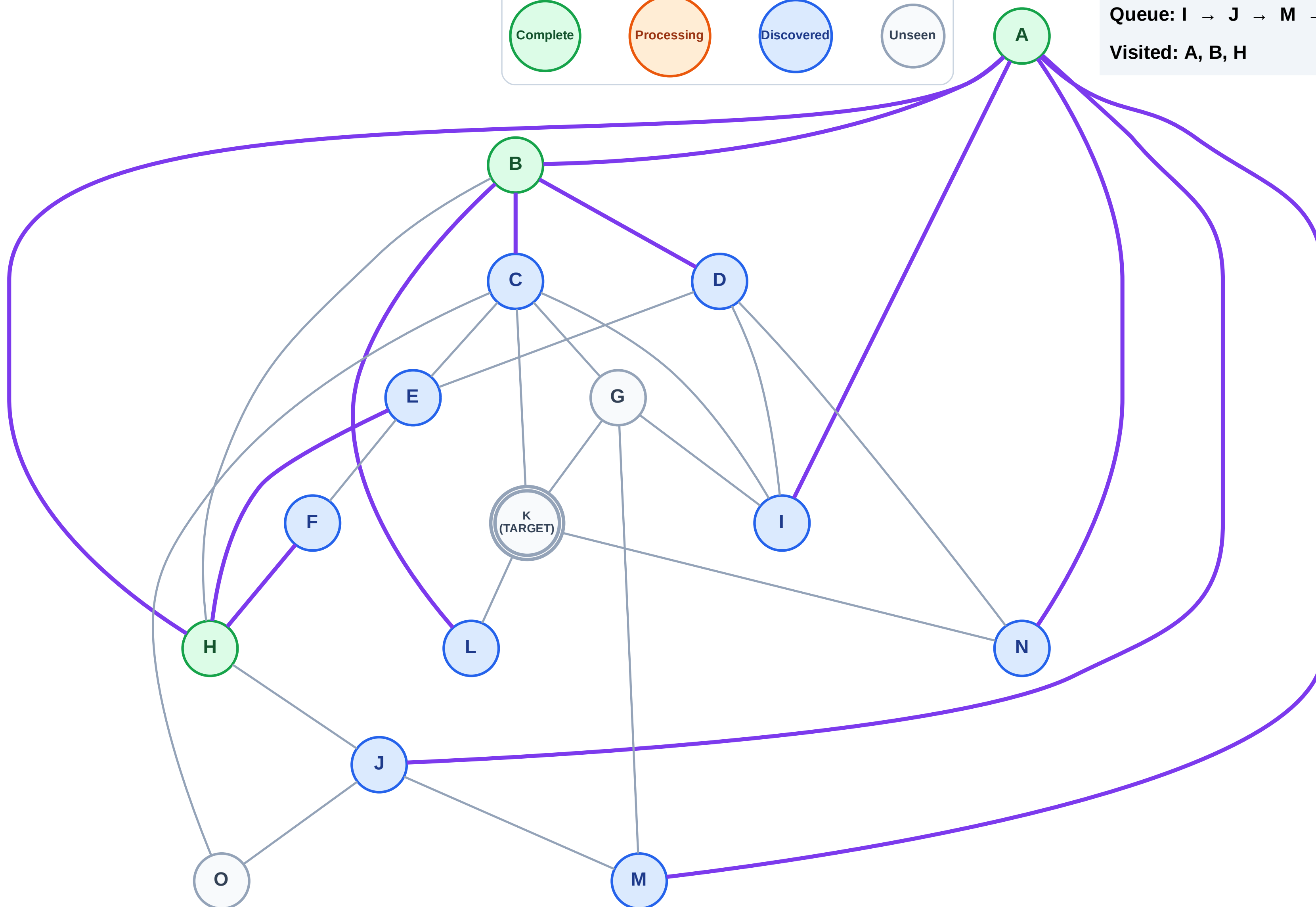
Discovered



Unseen

Queue: I → J → M → N → C → D → L → E → F

Visited: A, B, H



BFS Traversal

Step 8: Process node I

Legend

Complete

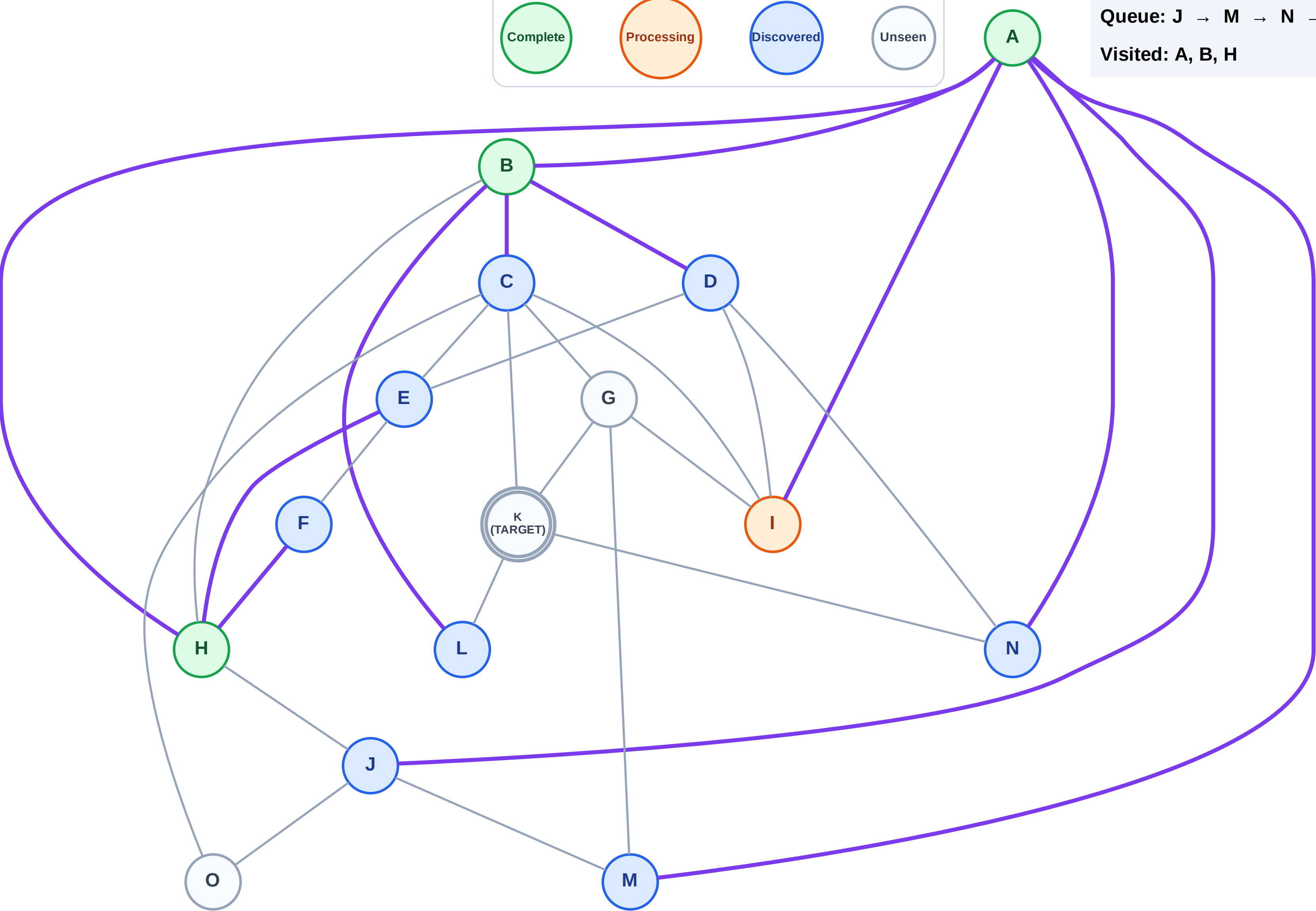
Processing

Discovered

Unseen

Queue: J → M → N → C → D → L → E → F

Visited: A, B, H



BFS Traversal

Step 9: Discover G from I

Legend

Complete

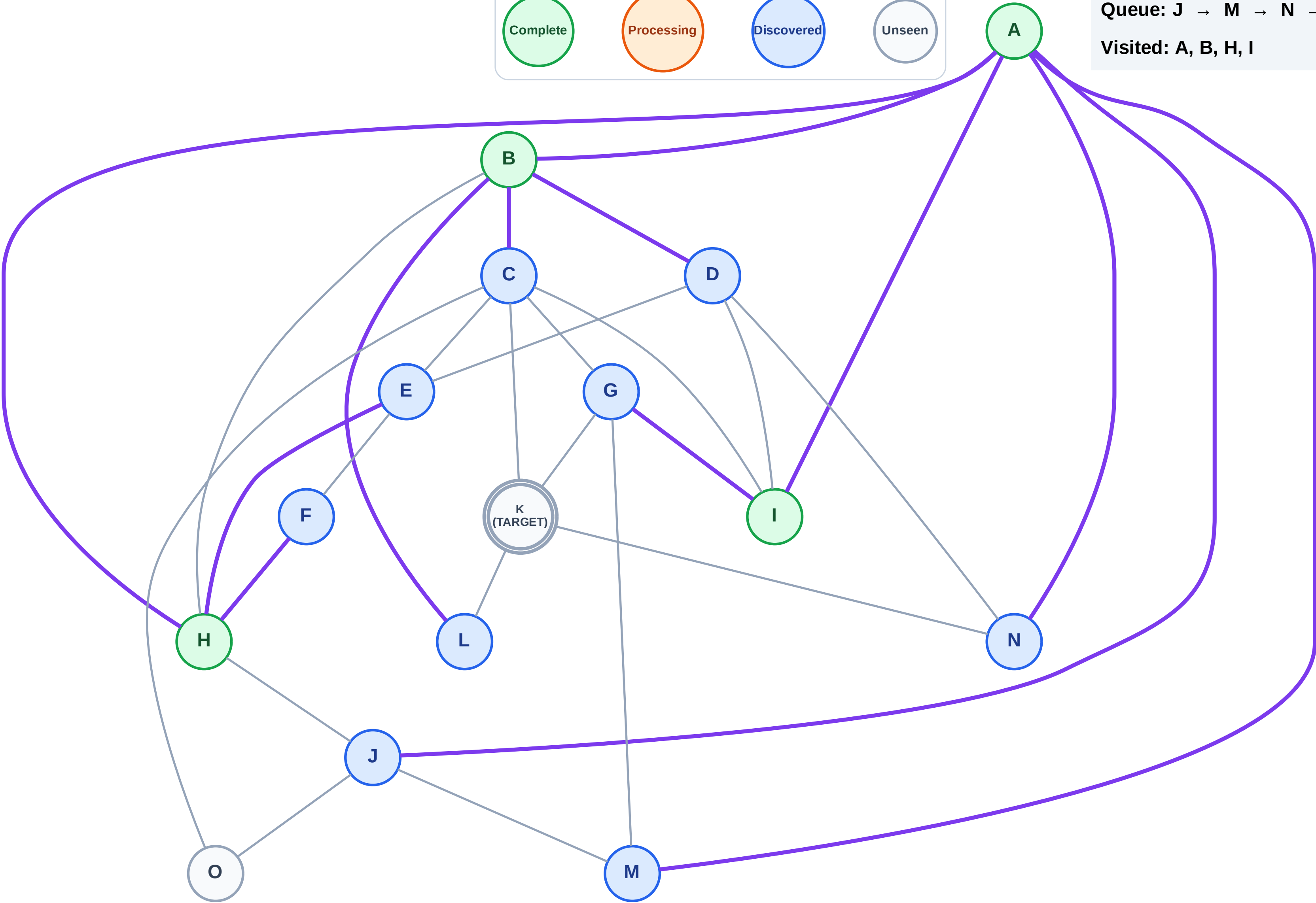
Processing

Discovered

Unseen

Queue: J → M → N → C → D → L → E → F → G

Visited: A, B, H, I



BFS Traversal

Step 10: Process node J

Legend

Complete

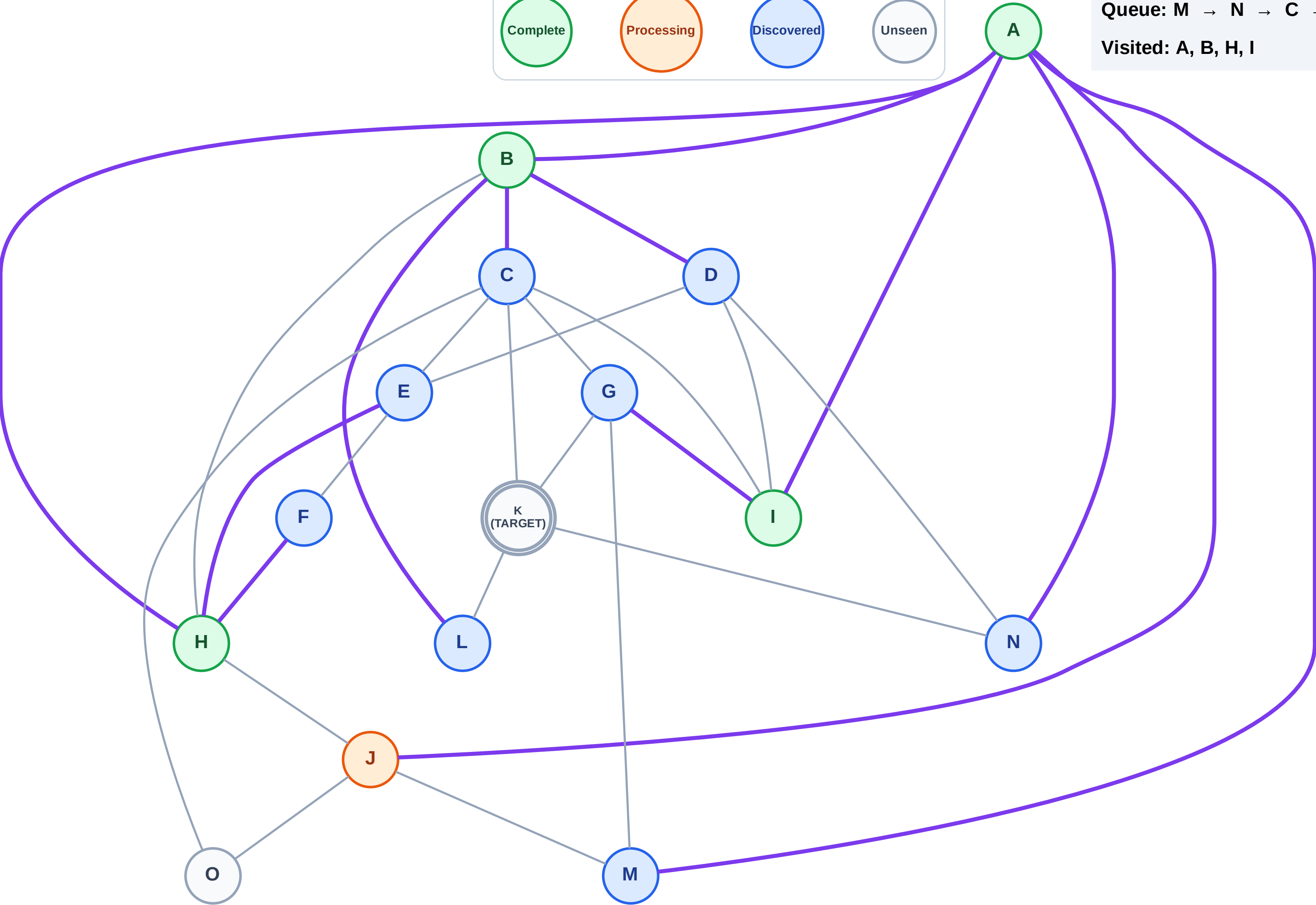
Processing

Discovered

Unseen

Queue: M → N → C → D → L → E → F → G

Visited: A, B, H, I



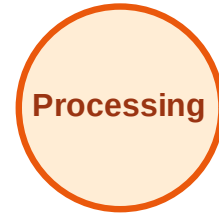
BFS Traversal

Step 11: Discover O from J

Legend



Complete



Processing



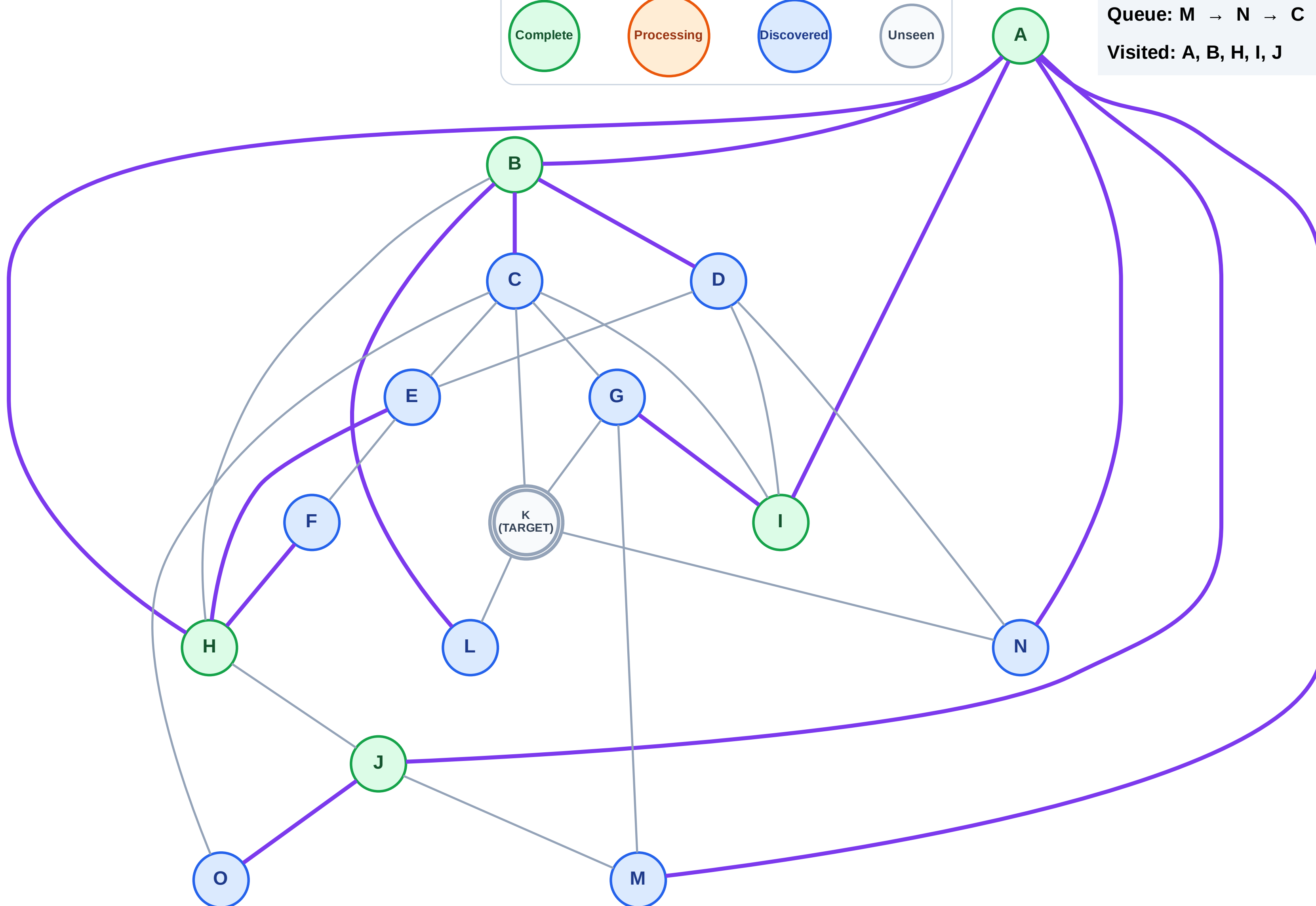
Discovered



Unseen

Queue: M → N → C → D → L → E → F → G → O

Visited: A, B, H, I, J



BFS Traversal

Step 12: Process node M

Legend

Complete

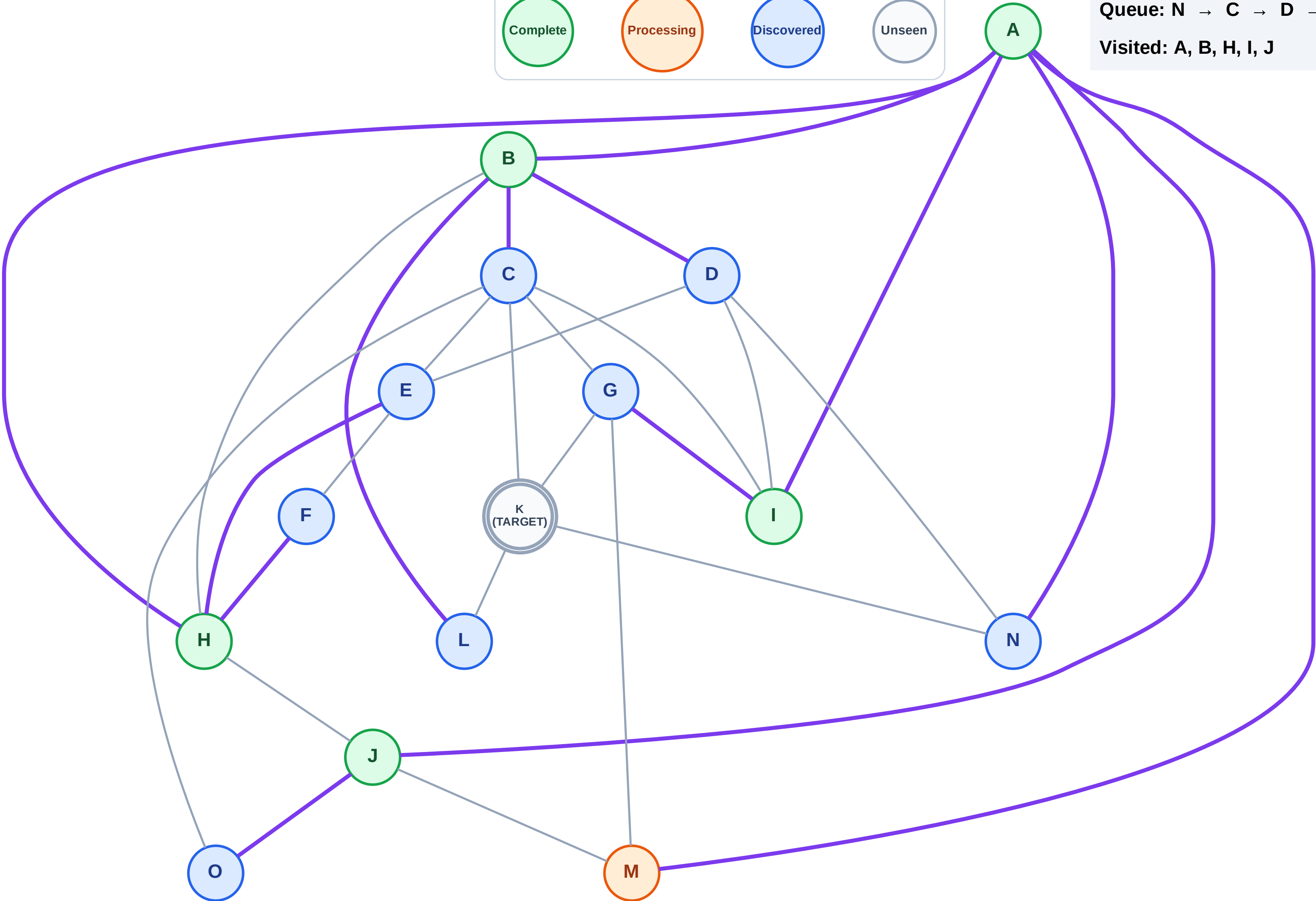
Processing

Discovered

Unseen

Queue: N → C → D → L → E → F → G → O

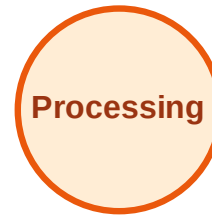
Visited: A, B, H, I, J



BFS Traversal

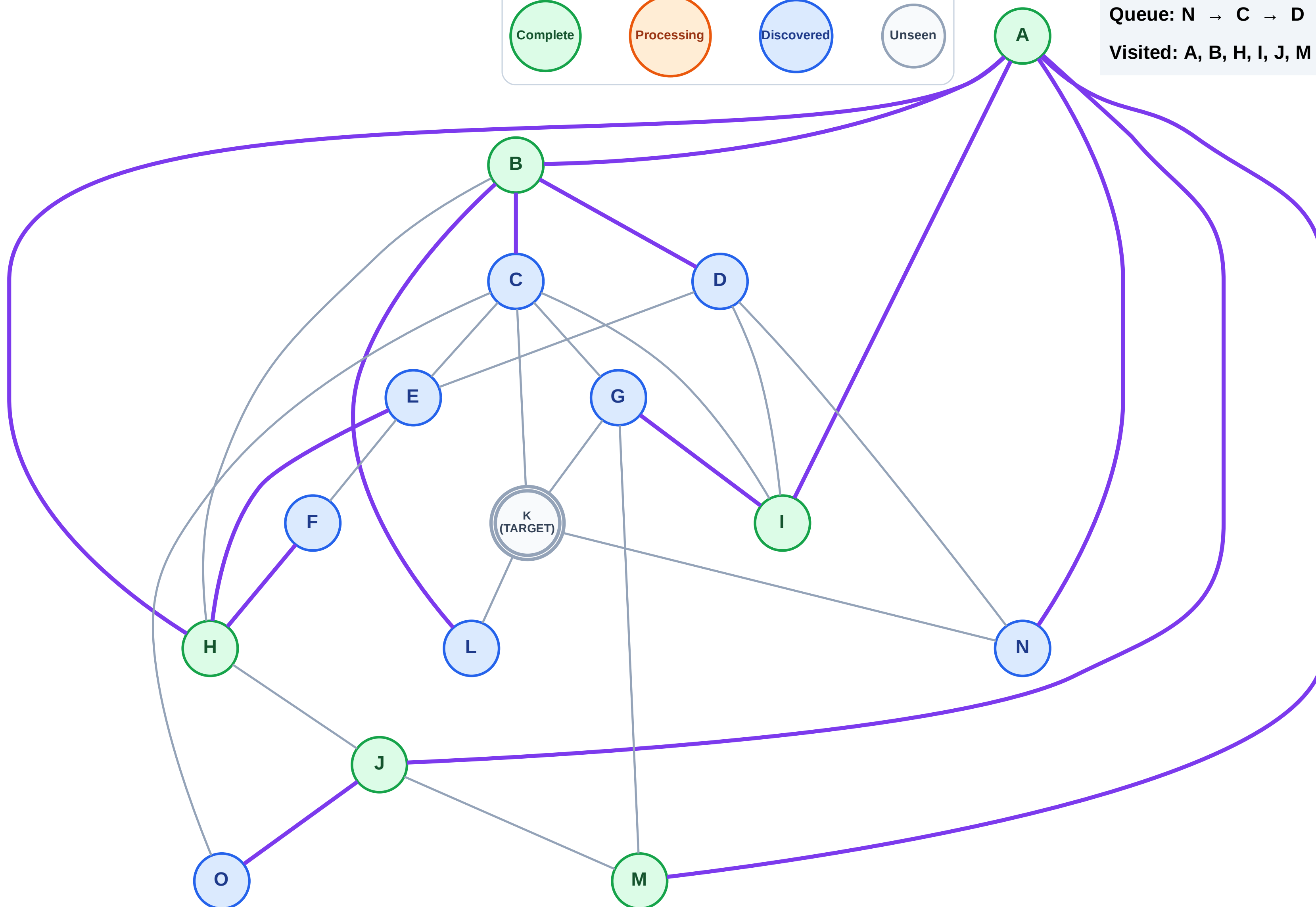
Step 13: No new neighbors from M

Legend



Queue: N → C → D → L → E → F → G → O

Visited: A, B, H, I, J, M



BFS Traversal

Step 14: Process node N

Legend

Complete

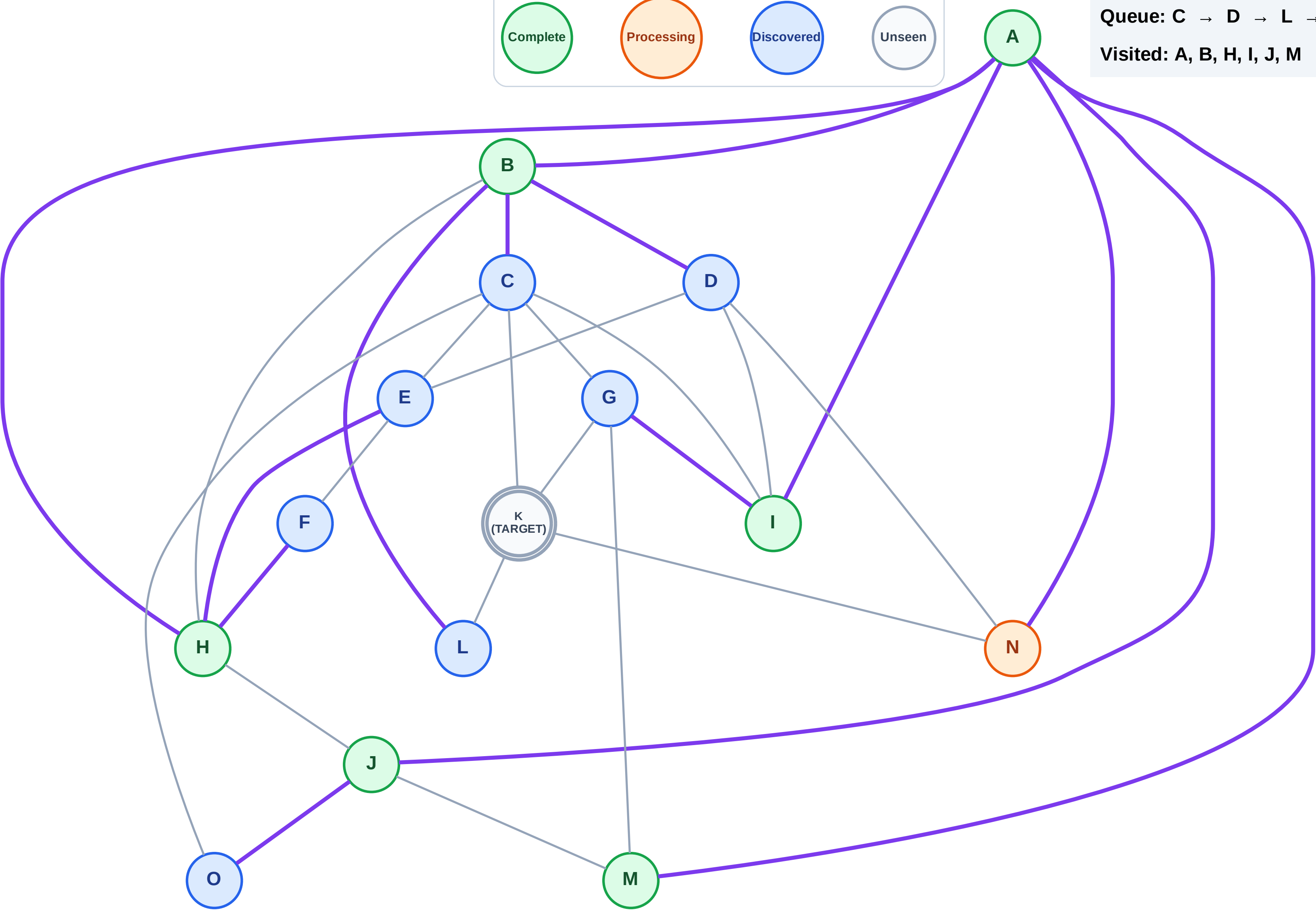
Processing

Discovered

Unseen

Queue: C → D → L → E → F → G → O

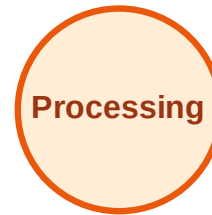
Visited: A, B, H, I, J, M



BFS Traversal

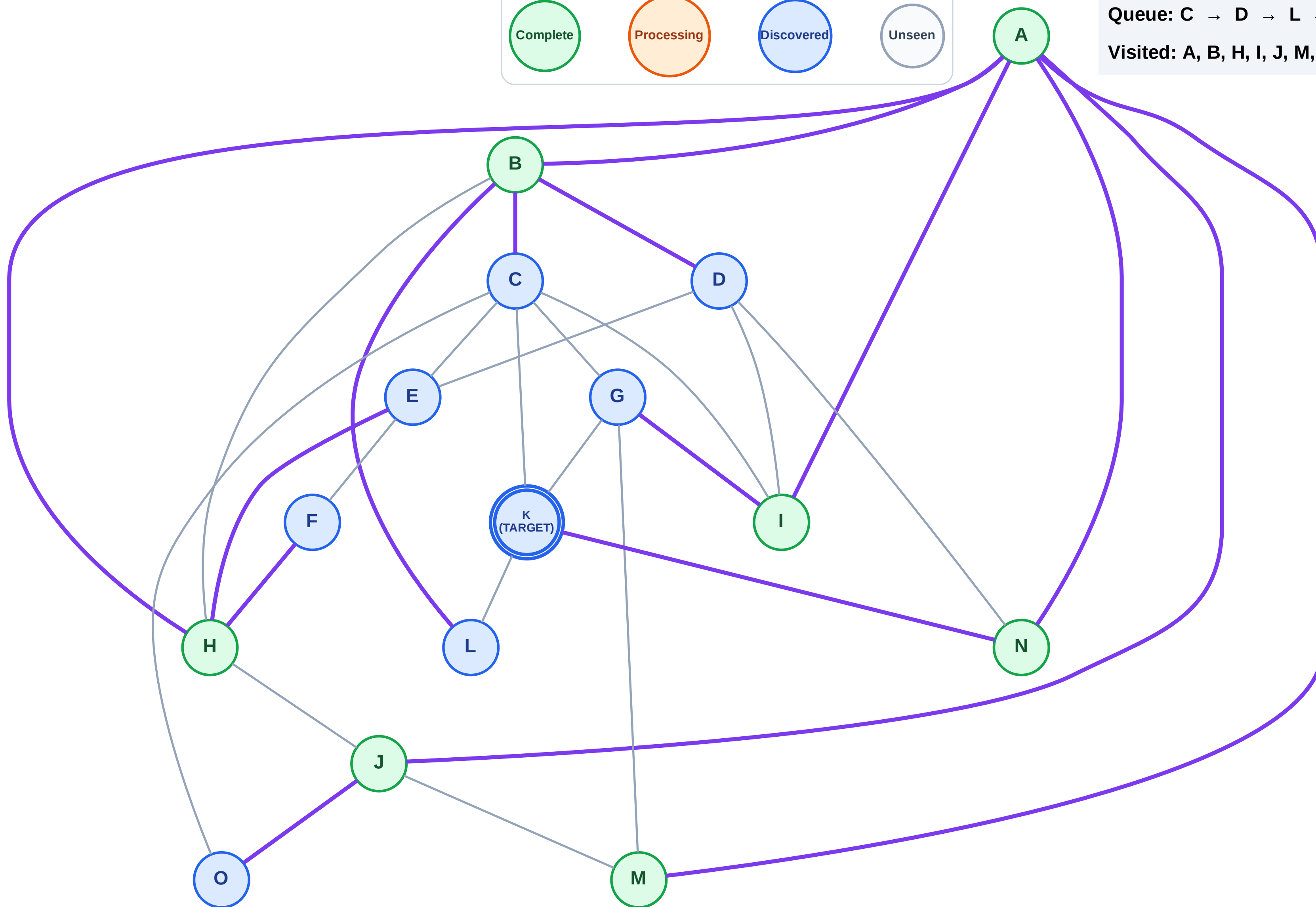
Step 15: Discover K from N

Legend



Queue: C → D → L → E → F → G → O → K

Visited: A, B, H, I, J, M, N



BFS Traversal

Step 16: Process node C

Legend

Complete

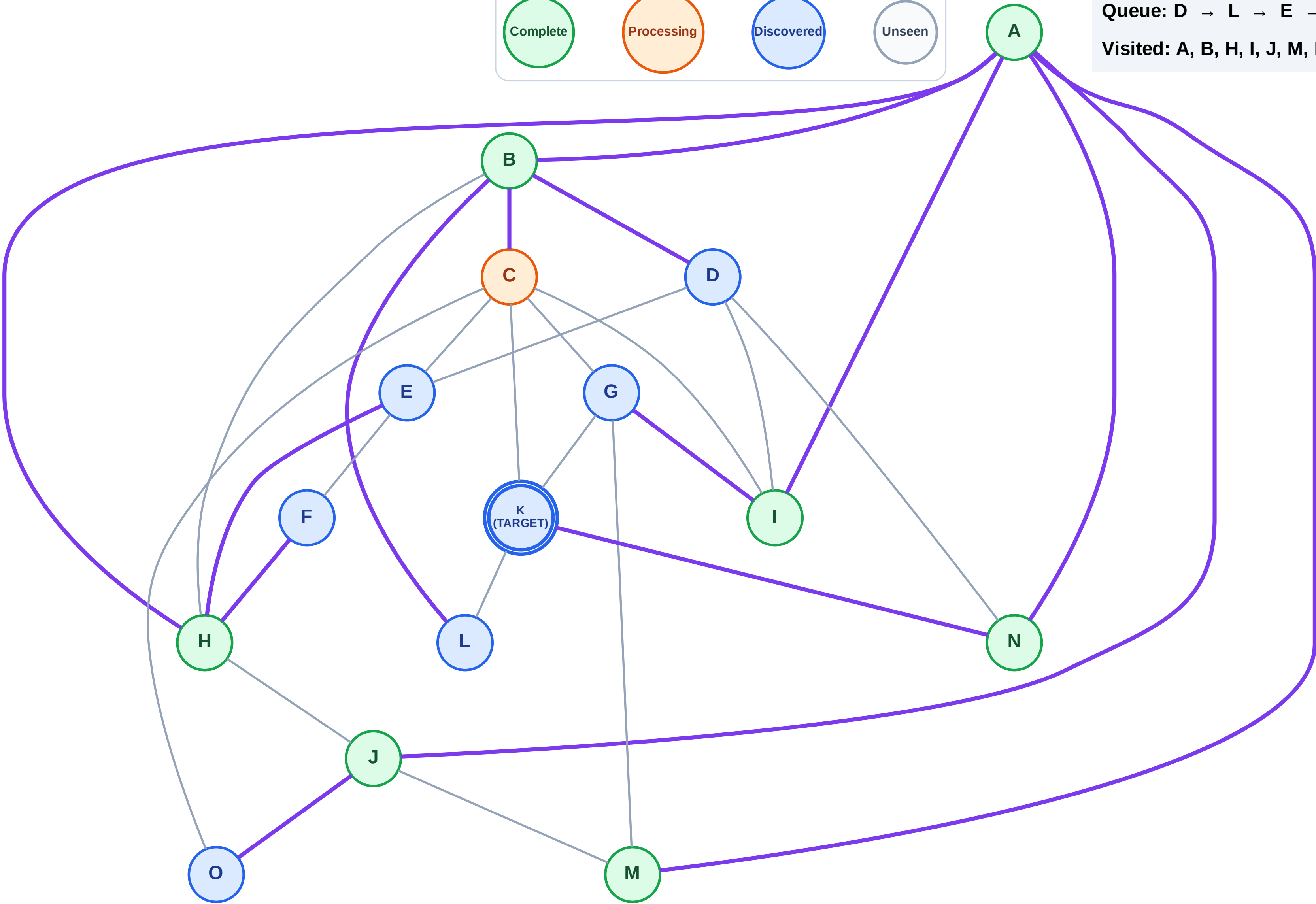
Processing

Discovered

Unseen

Queue: D → L → E → F → G → O → K

Visited: A, B, H, I, J, M, N



BFS Traversal

Step 17: No new neighbors from C

Legend

Complete

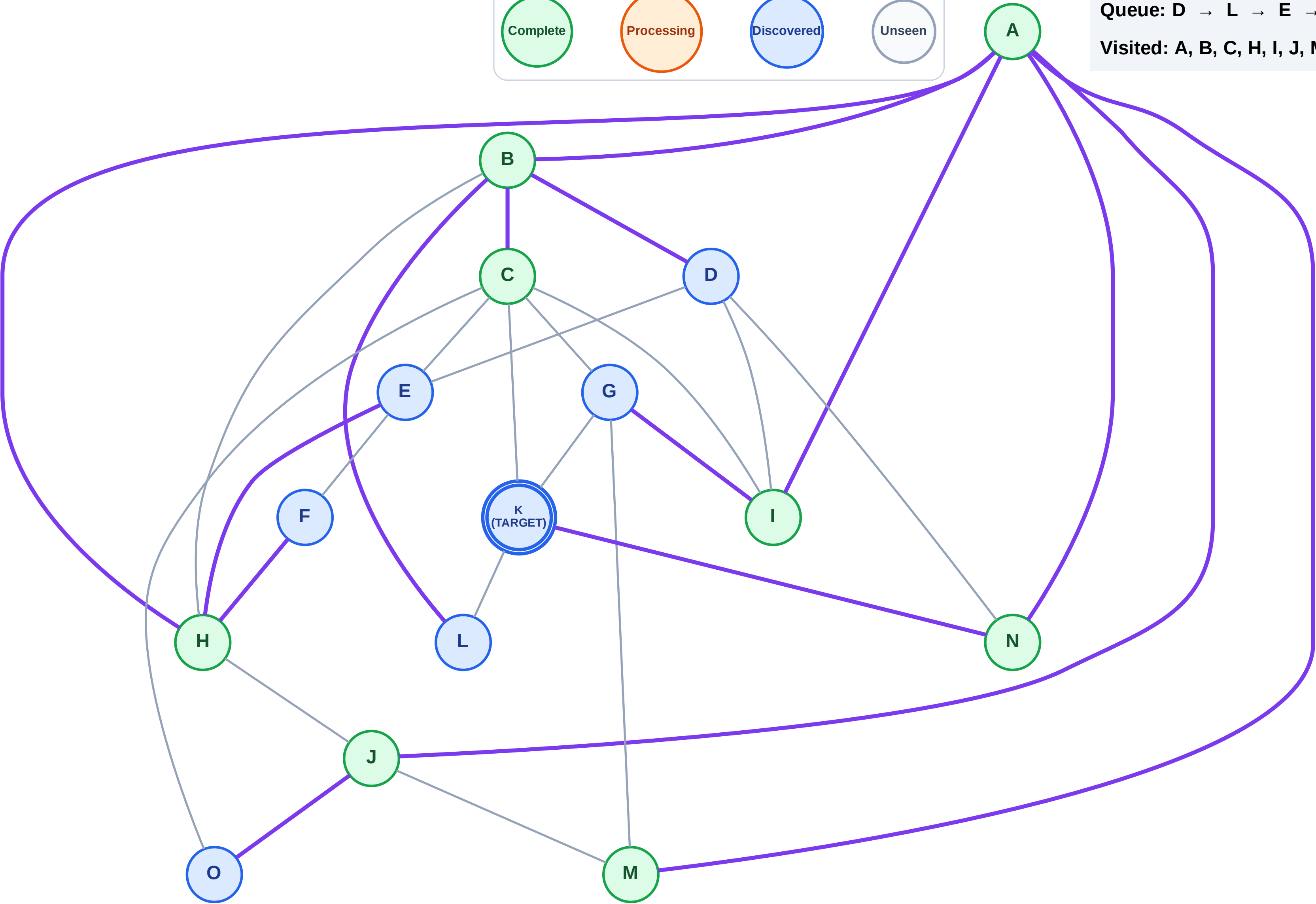
Processing

Discovered

Unseen

Queue: D → L → E → F → G → O → K

Visited: A, B, C, H, I, J, M, N



Step 18: Process node D

Legend



Complete



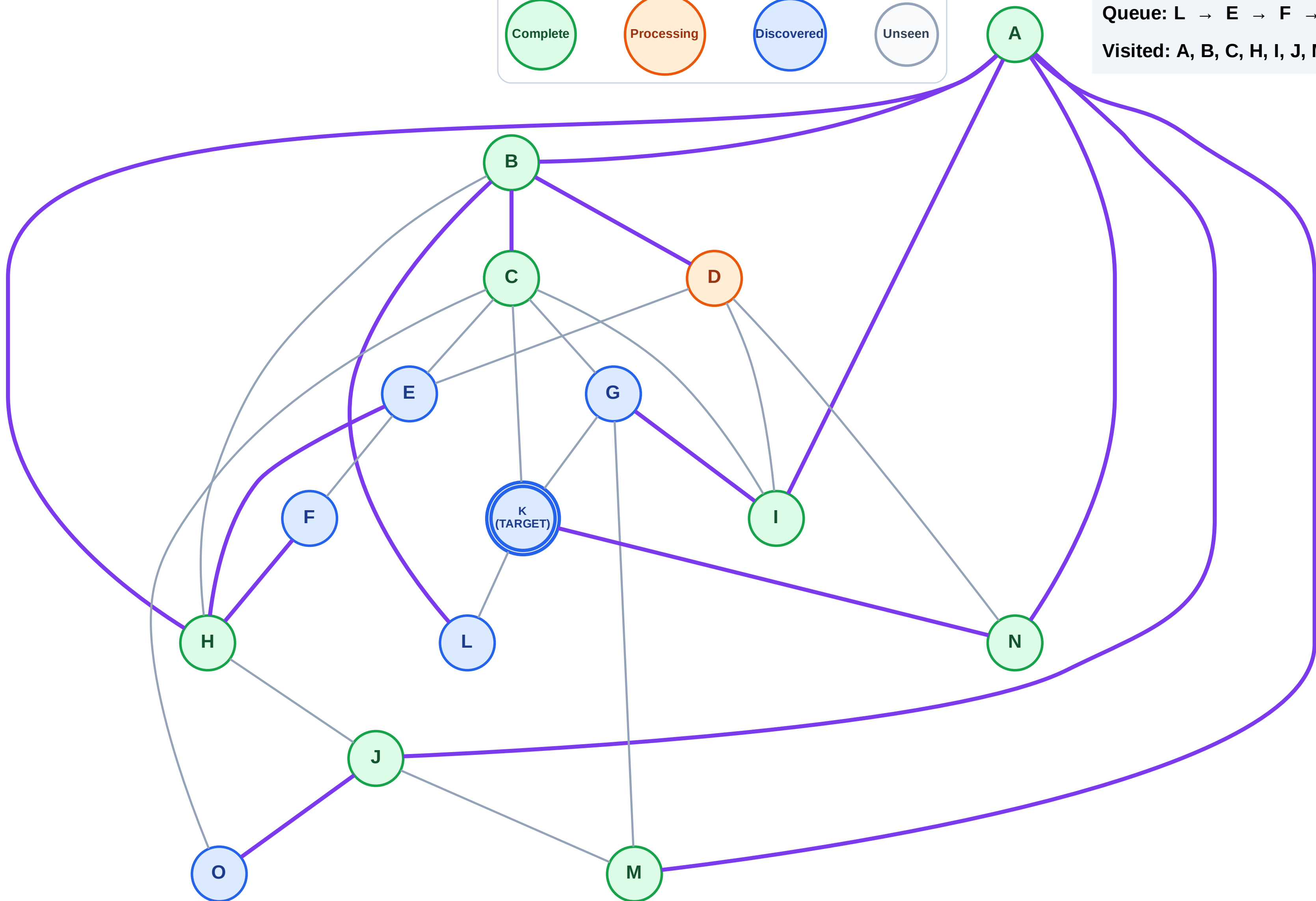
Processing



Discovered



Visited: A, B, C, H, I, J, M, N



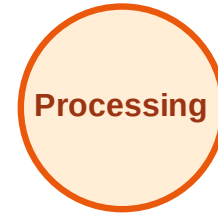
BFS Traversal

Step 19: No new neighbors from D

Legend



Complete



Processing



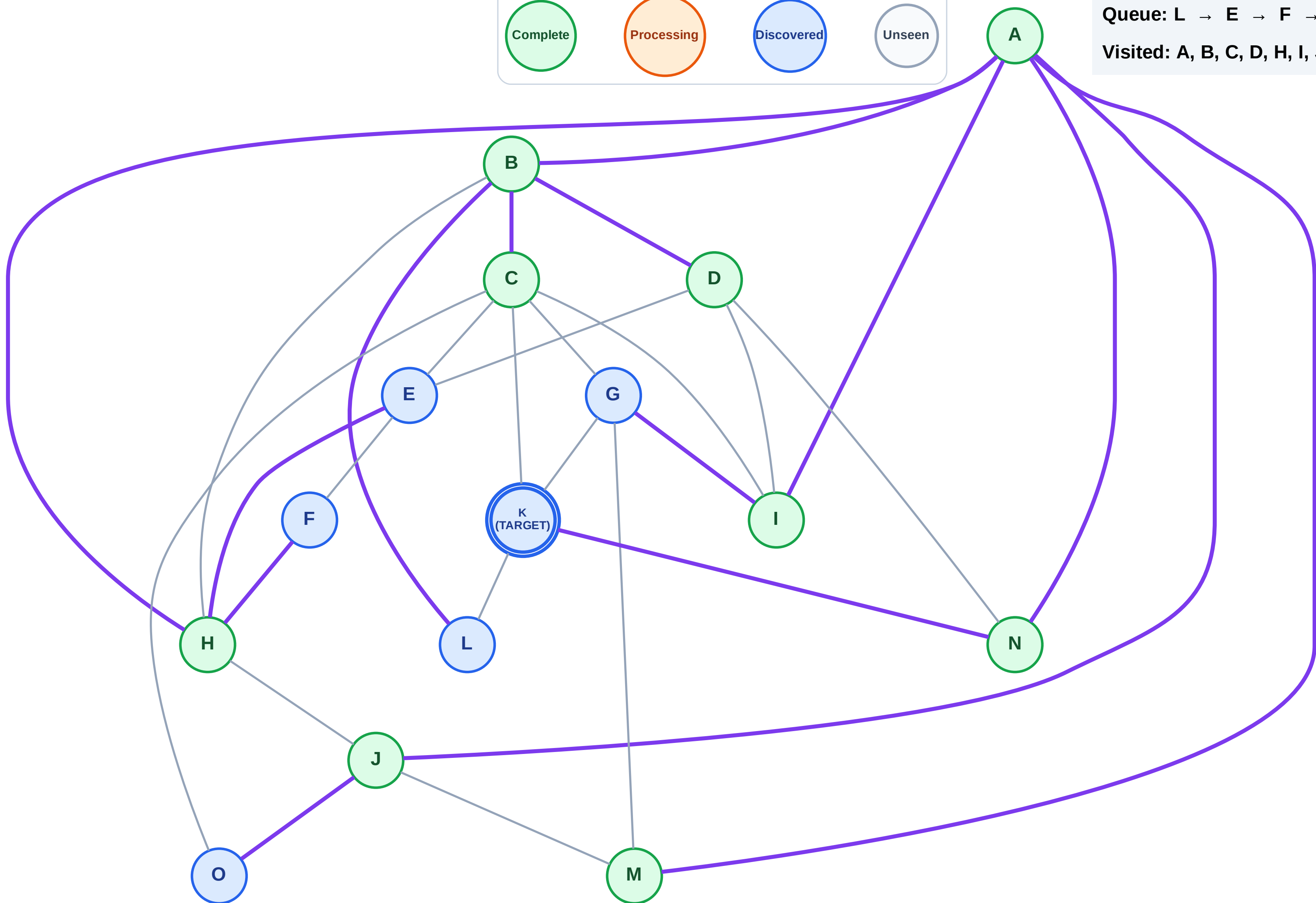
Discovered



Unseen

Queue: L → E → F → G → O → K

Visited: A, B, C, D, H, I, J, M, N



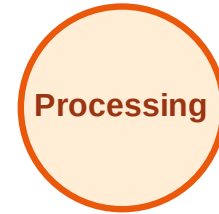
BFS Traversal

Step 20: Process node L

Legend



Complete



Processing



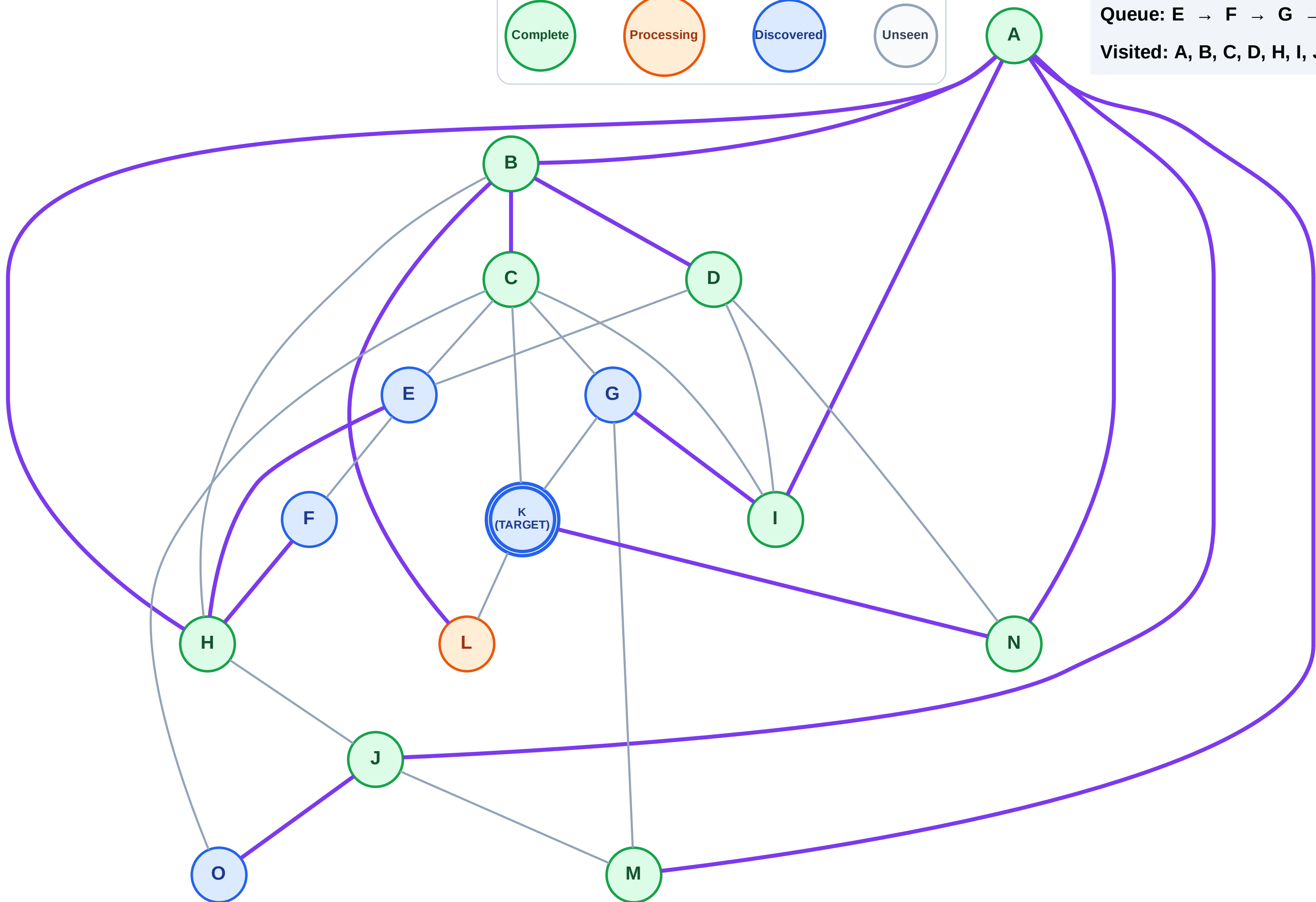
Discovered



Unseen

Queue: E → F → G → O → K

Visited: A, B, C, D, H, I, J, M, N



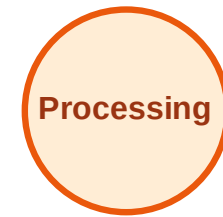
BFS Traversal

Step 21: No new neighbors from L

Legend



Complete



Processing



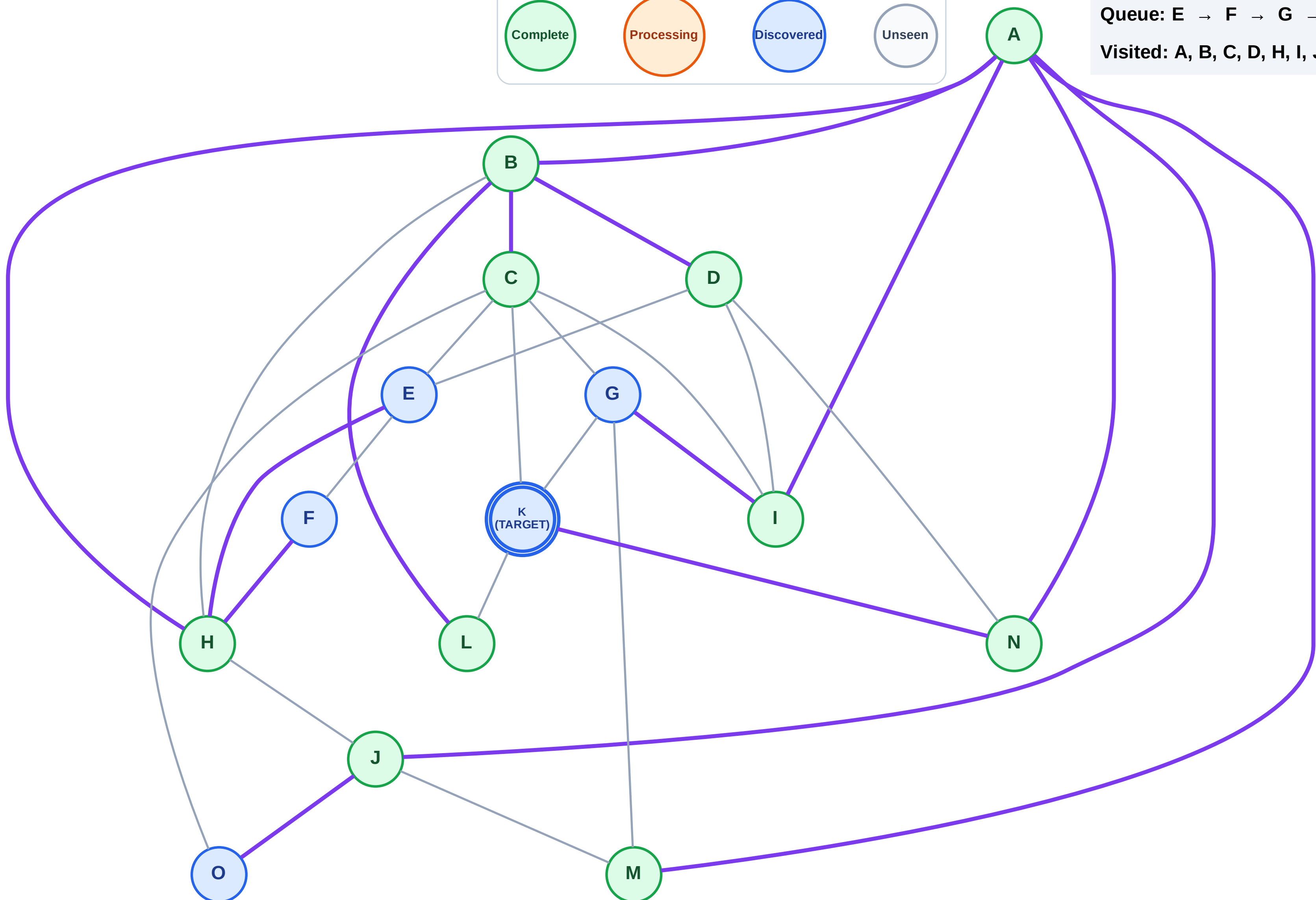
Discovered



Unseen

Queue: E → F → G → O → K

Visited: A, B, C, D, H, I, J, L, M, N



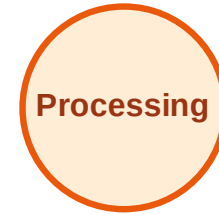
BFS Traversal

Step 22: Process node E

Legend



Complete



Processing



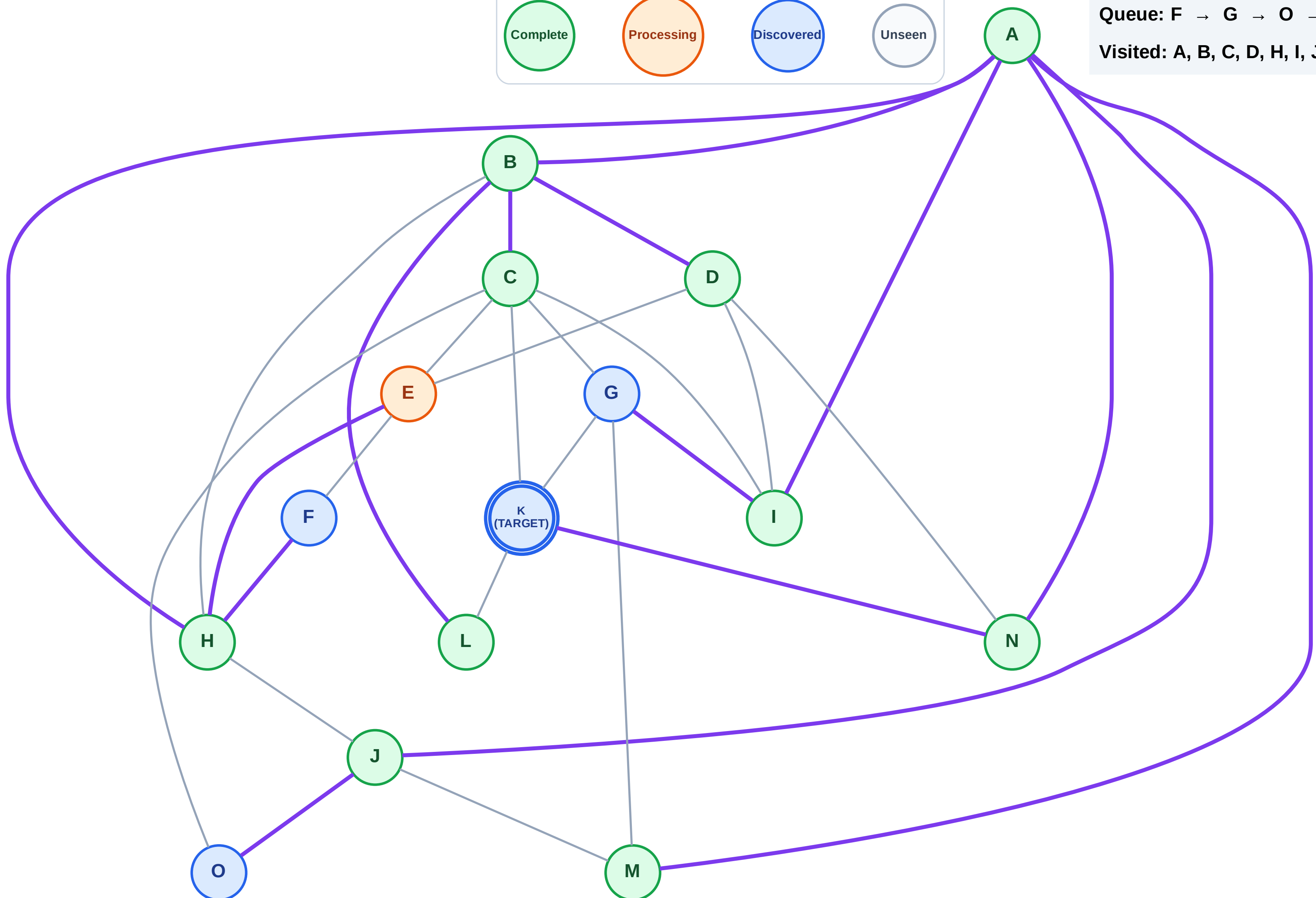
Discovered



Unseen

Queue: F → G → O → K

Visited: A, B, C, D, H, I, J, L, M, N



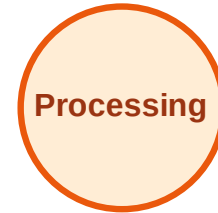
BFS Traversal

Step 23: No new neighbors from E

Legend



Complete



Processing



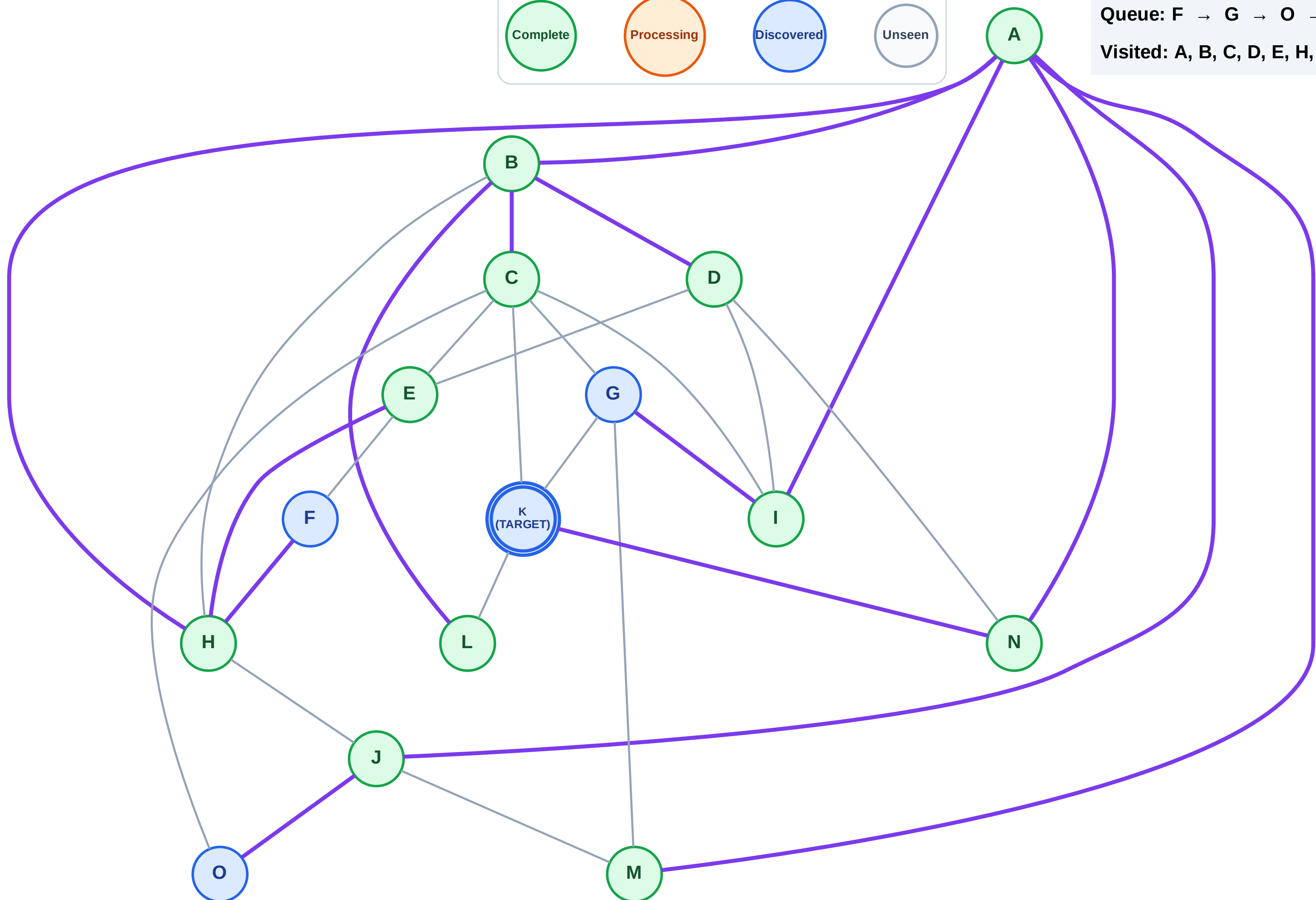
Discovered



Unseen

Queue: F → G → O → K

Visited: A, B, C, D, E, H, I, J, L, M, N



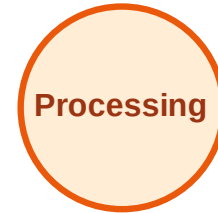
BFS Traversal

Step 24: Process node F

Legend



Complete



Processing



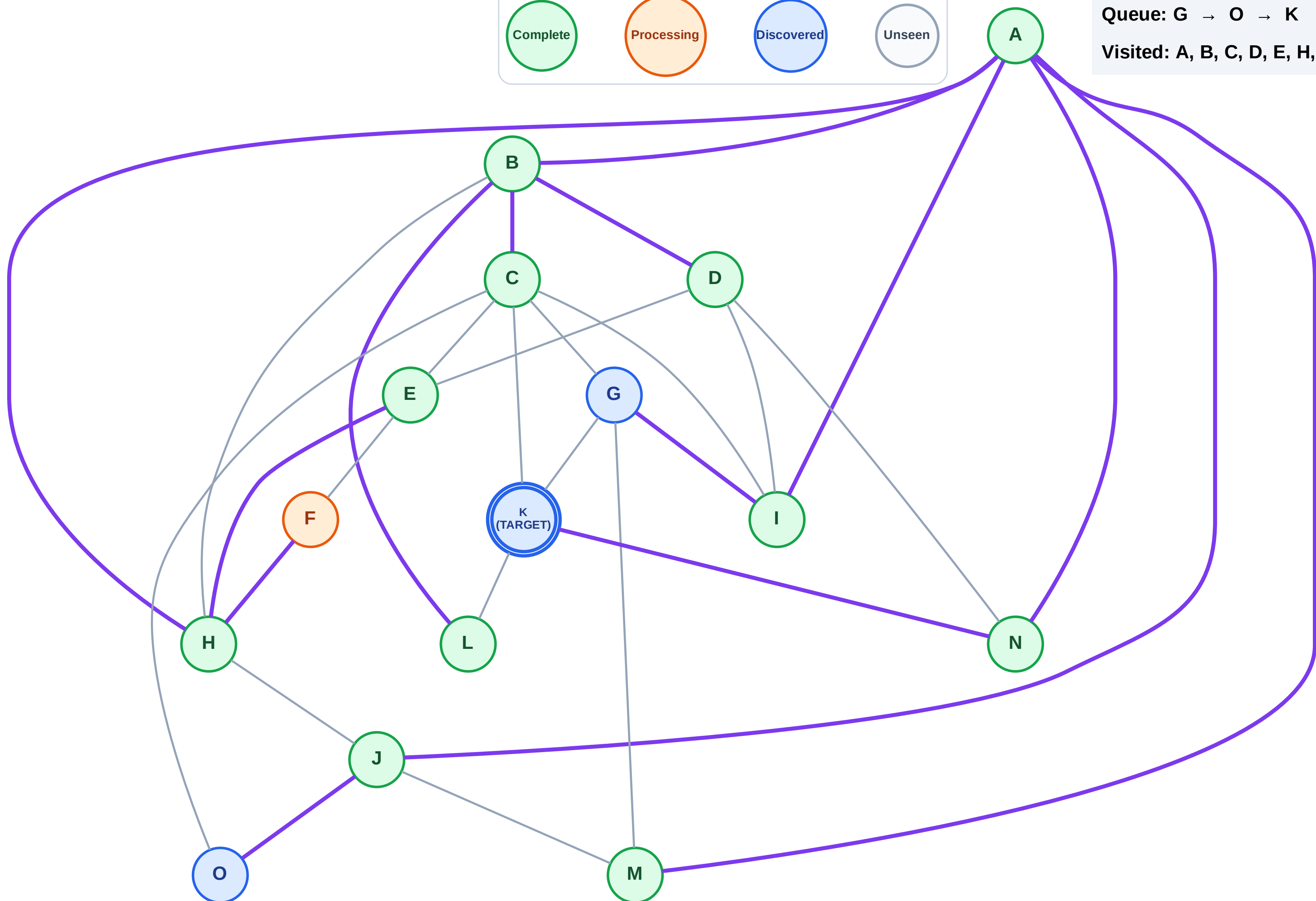
Discovered



Unseen

Queue: G → O → K

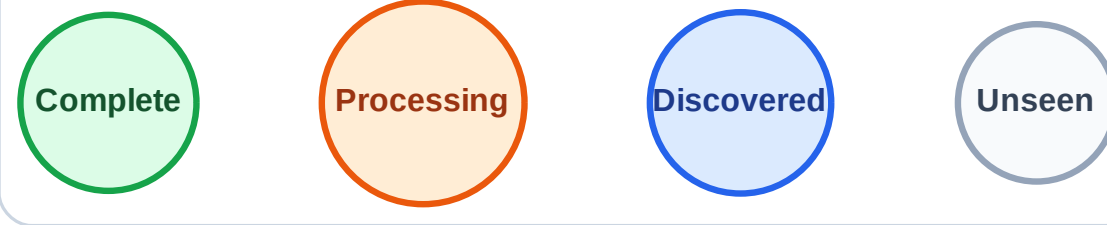
Visited: A, B, C, D, E, H, I, J, L, M, N



BFS Traversal

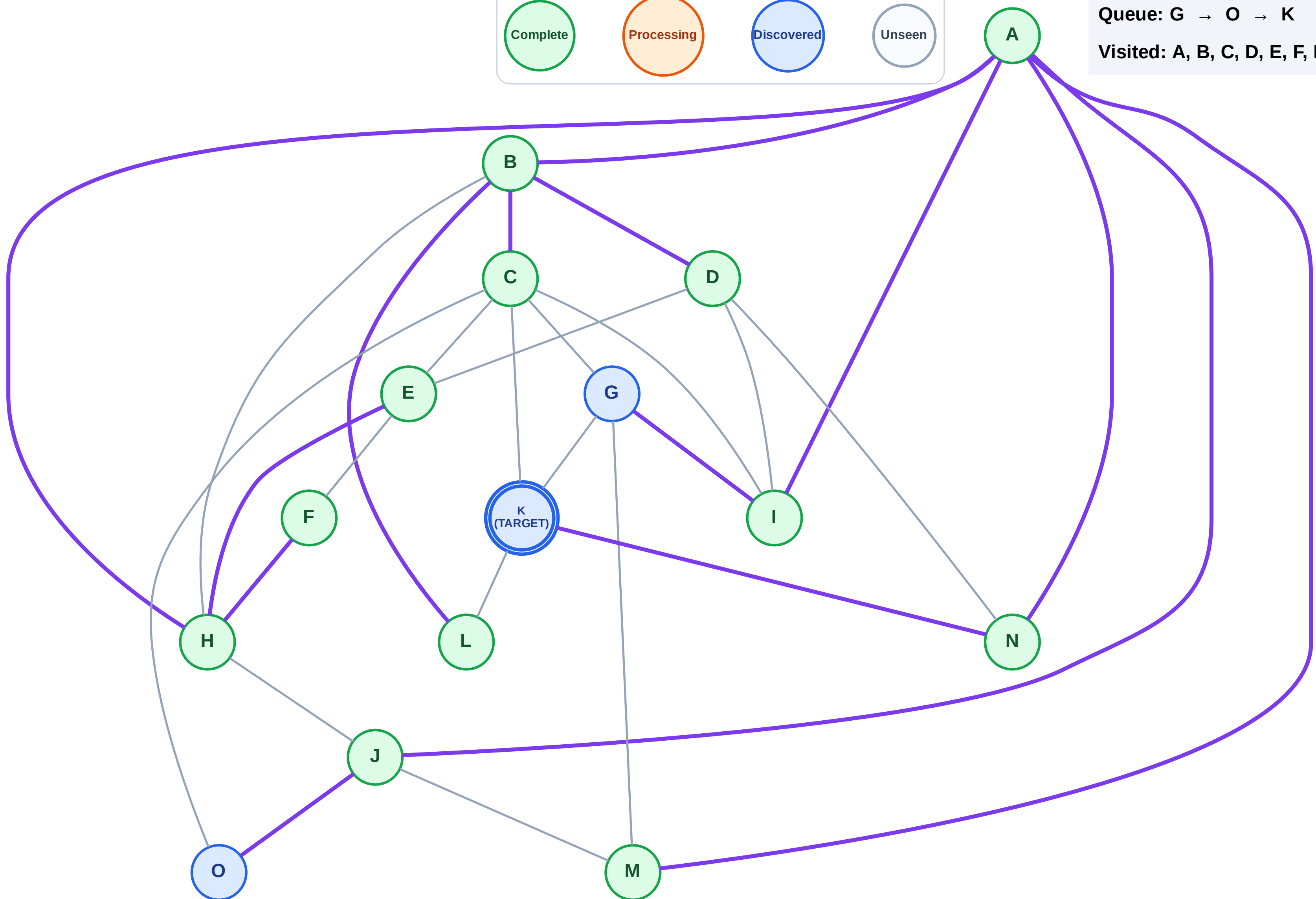
Step 25: No new neighbors from F

Legend



Queue: G → O → K

Visited: A, B, C, D, E, F, H, I, J, L, M, N



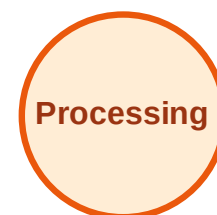
BFS Traversal

Step 26: Process node G

Legend



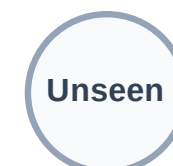
Complete



Processing



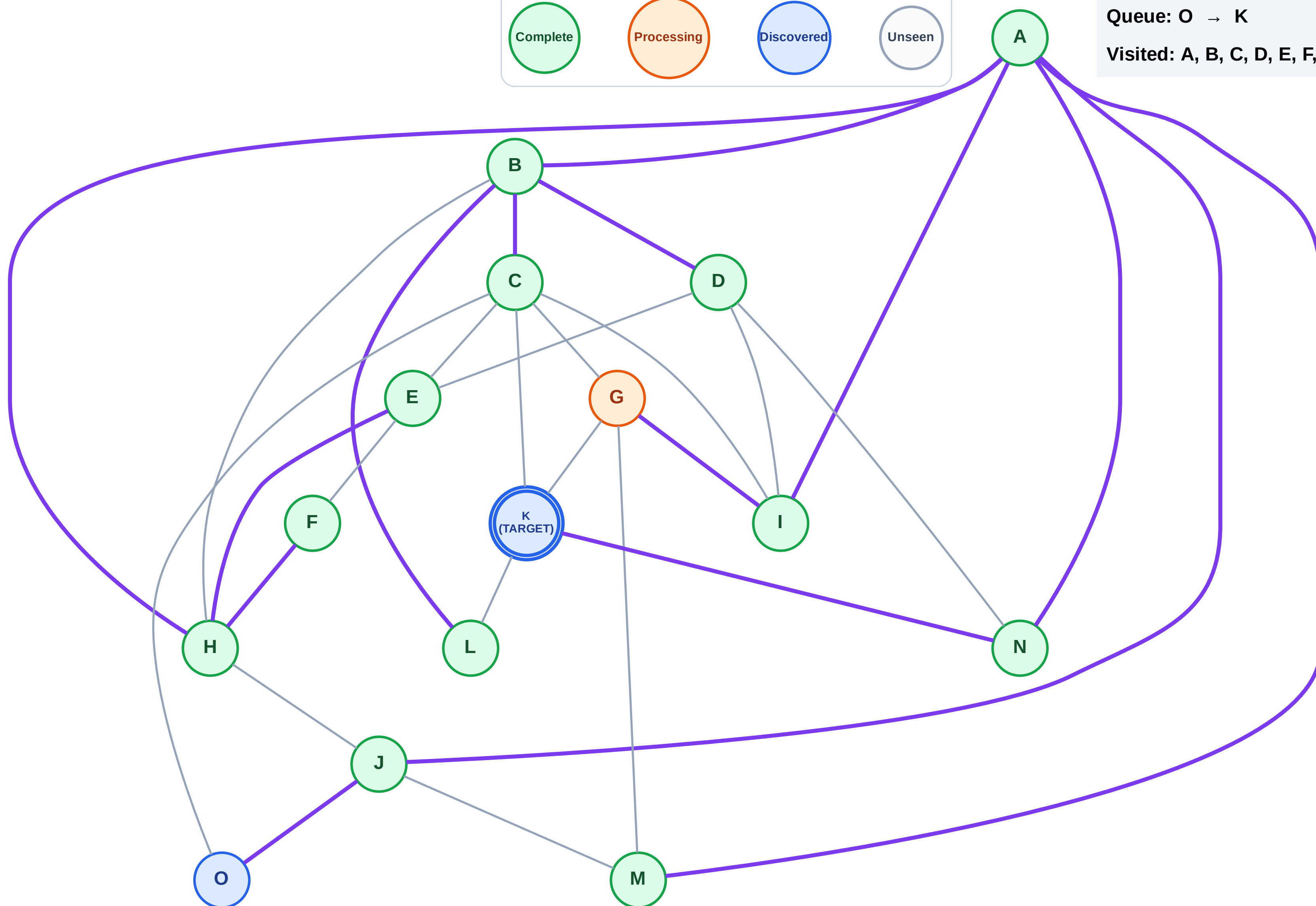
Discovered



Unseen

Queue: O → K

Visited: A, B, C, D, E, F, H, I, J, L, M, N



Step 27: No new neighbors from G

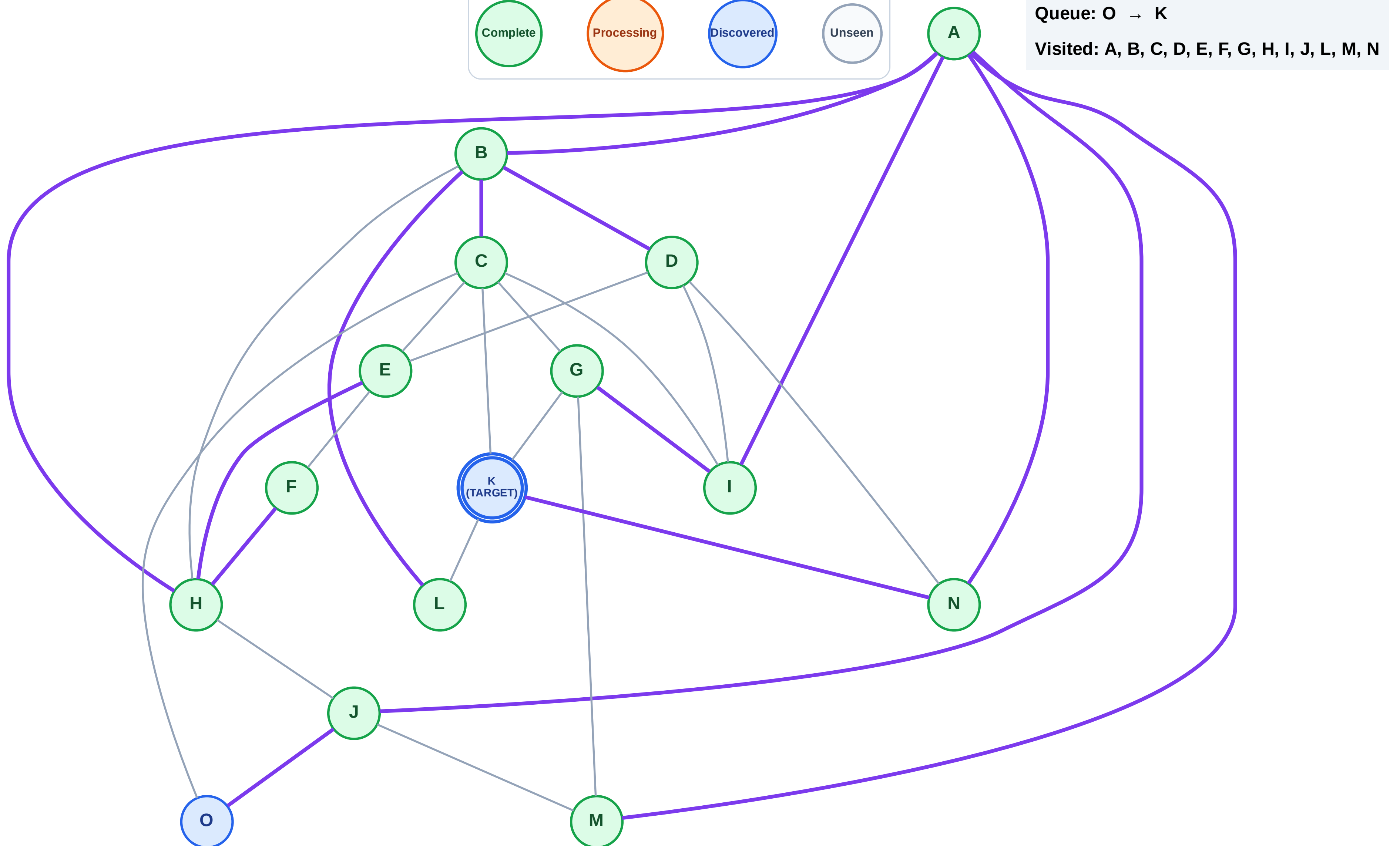
Step 27: No new neighbors from G

Legend



The legend consists of four colored circles arranged horizontally, each containing a label. From left to right: a green circle labeled 'Complete', an orange circle labeled 'Processing', a blue circle labeled 'Discovered', and a grey circle labeled 'Unseen'.

Visited: A, B, C, D, E, F, G, H, I, J, L, M, N



Step 28: Process node O

Step 28: Process node O

Legend

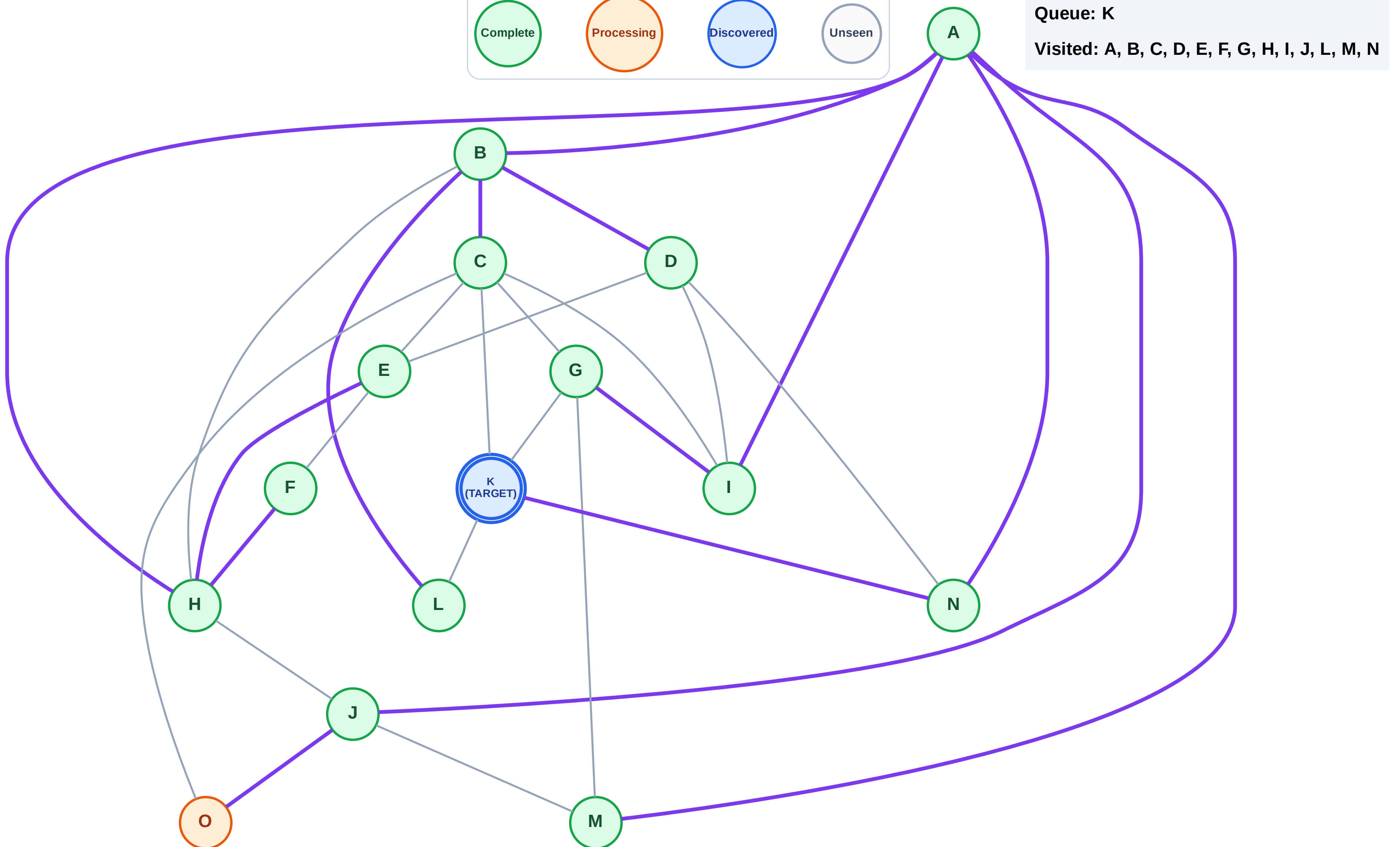


The legend consists of four colored circles arranged horizontally, each containing a label. From left to right: a green circle labeled 'Complete', an orange circle labeled 'Processing', a blue circle labeled 'Discovered', and a grey circle labeled 'Unseen'.

Processing

Unseen

Visited: A, B, C, D, E, F, G, H, I, J, L, M, N



BFS Traversal

Step 29: No new neighbors from O

Legend



Complete



Processing



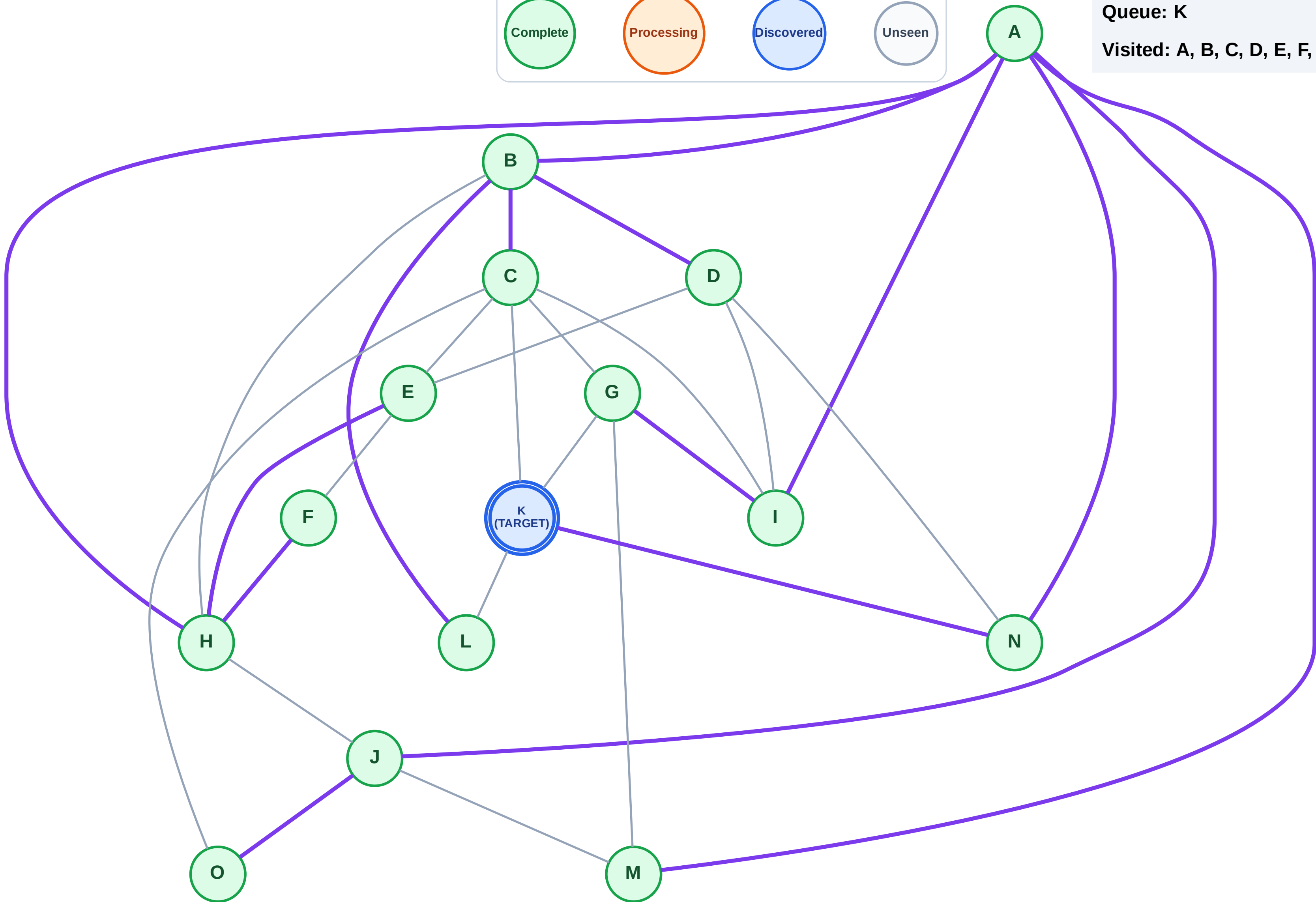
Discovered



Unseen

Queue: K

Visited: A, B, C, D, E, F, G, H, I, J, L, M, N, O



BFS Traversal

Step 30: Process node K

Legend

Complete

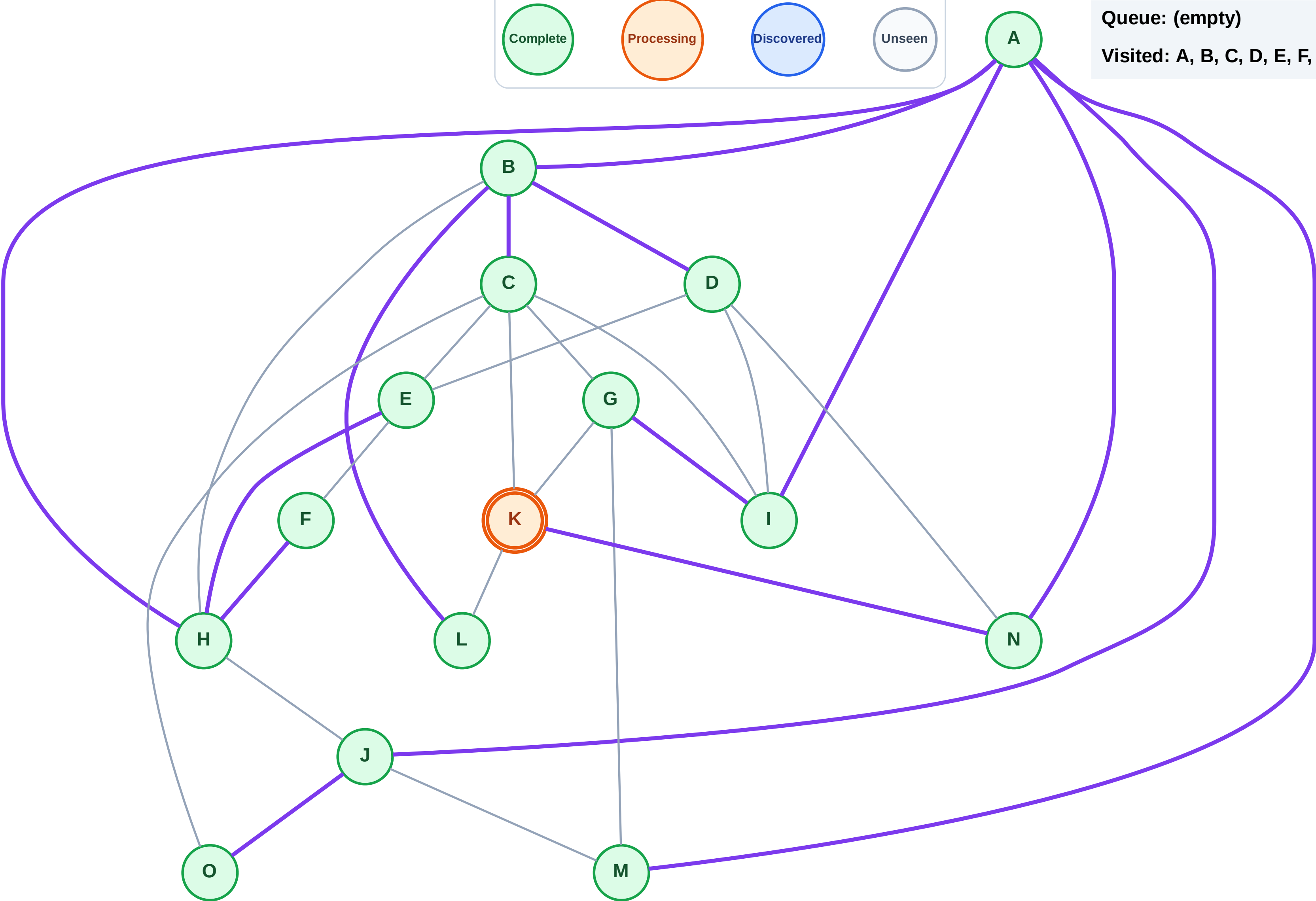
Processing

Discovered

Unseen

Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J, L, M, N, O



BFS Traversal

Step 31: TARGET NODE K REACHED! Path (2 hops): A → N → K

Legend

Complete

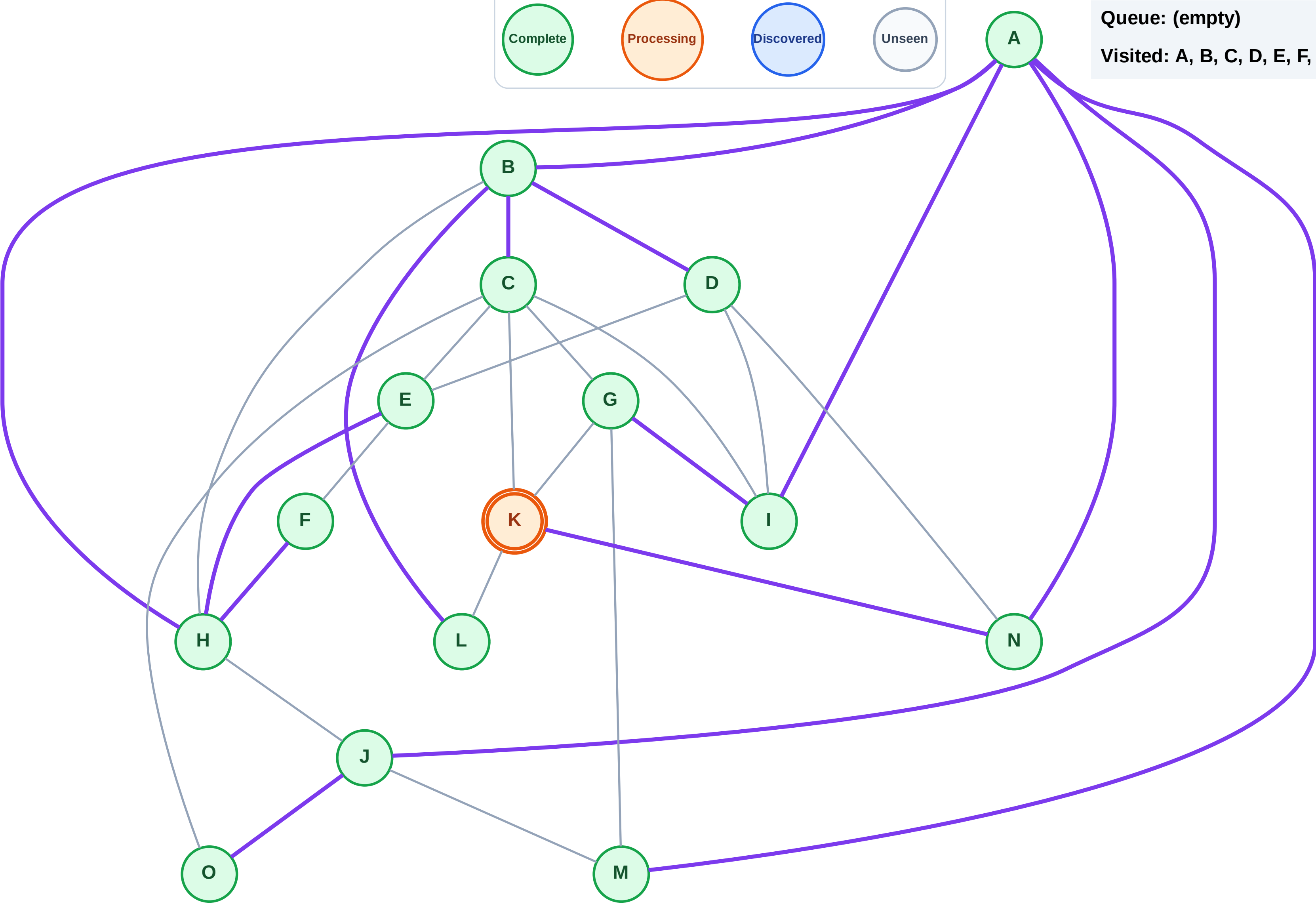
Processing

Discovered

Unseen

Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, N, O



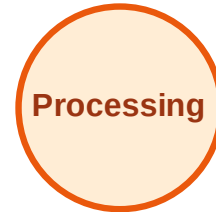
DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Complete



Processing



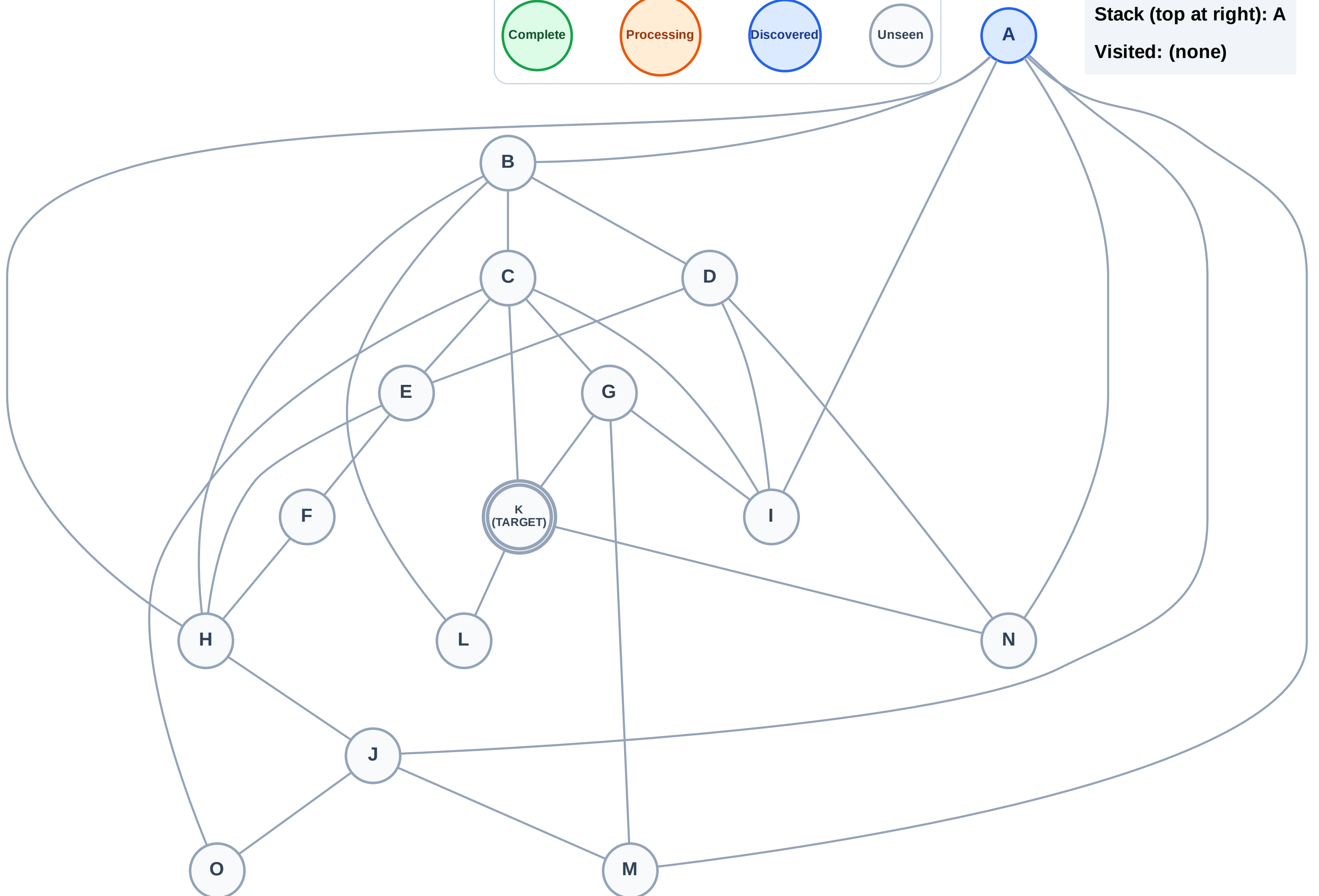
Discovered



Unseen

Stack (top at right): A

Visited: (none)



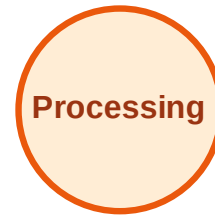
DFS Traversal

Step 2: Process node A

Legend



Complete



Processing



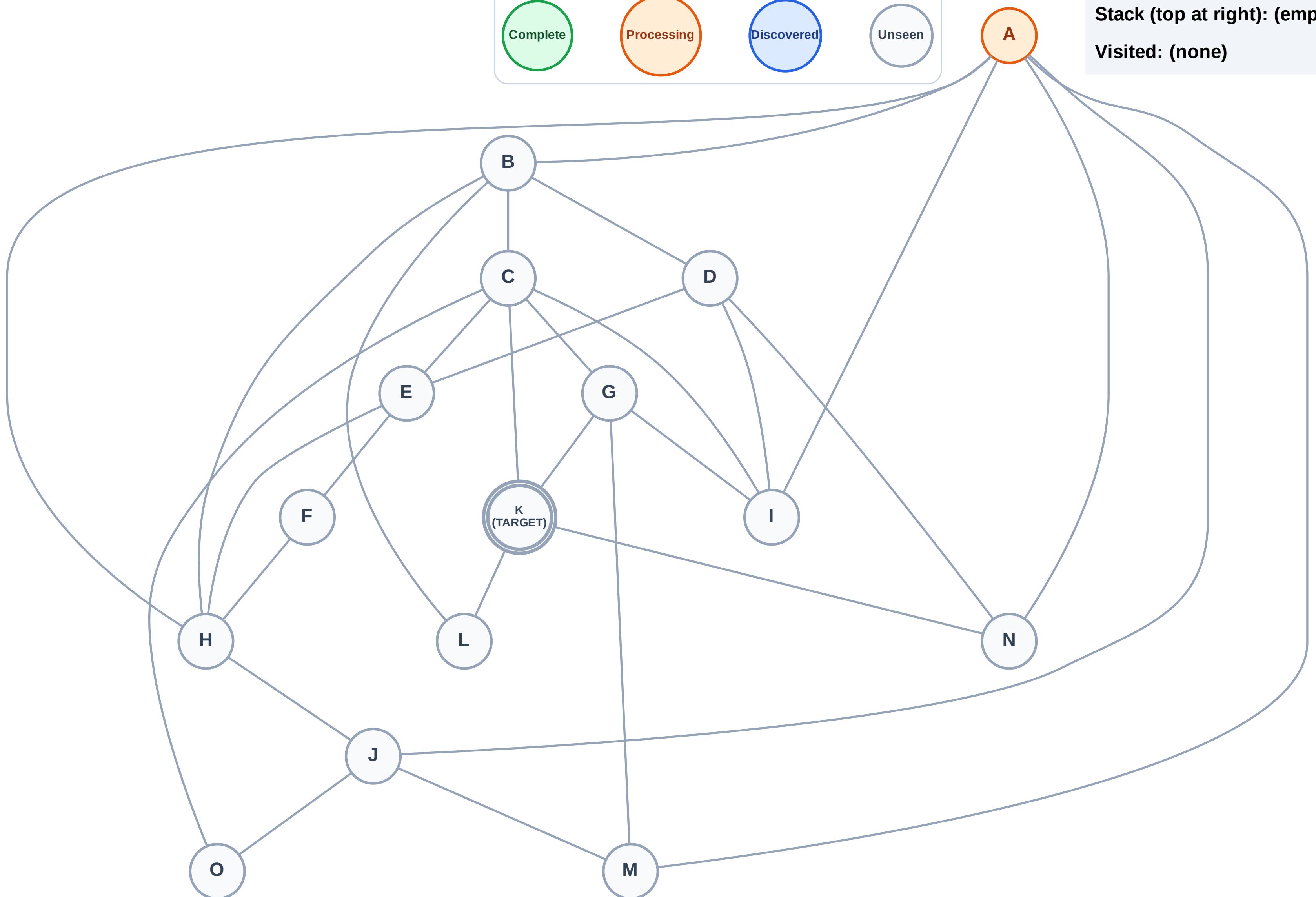
Discovered



Unseen

Stack (top at right): (empty)

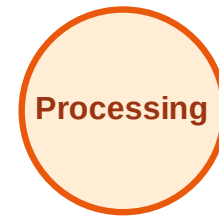
Visited: (none)



DFS Traversal

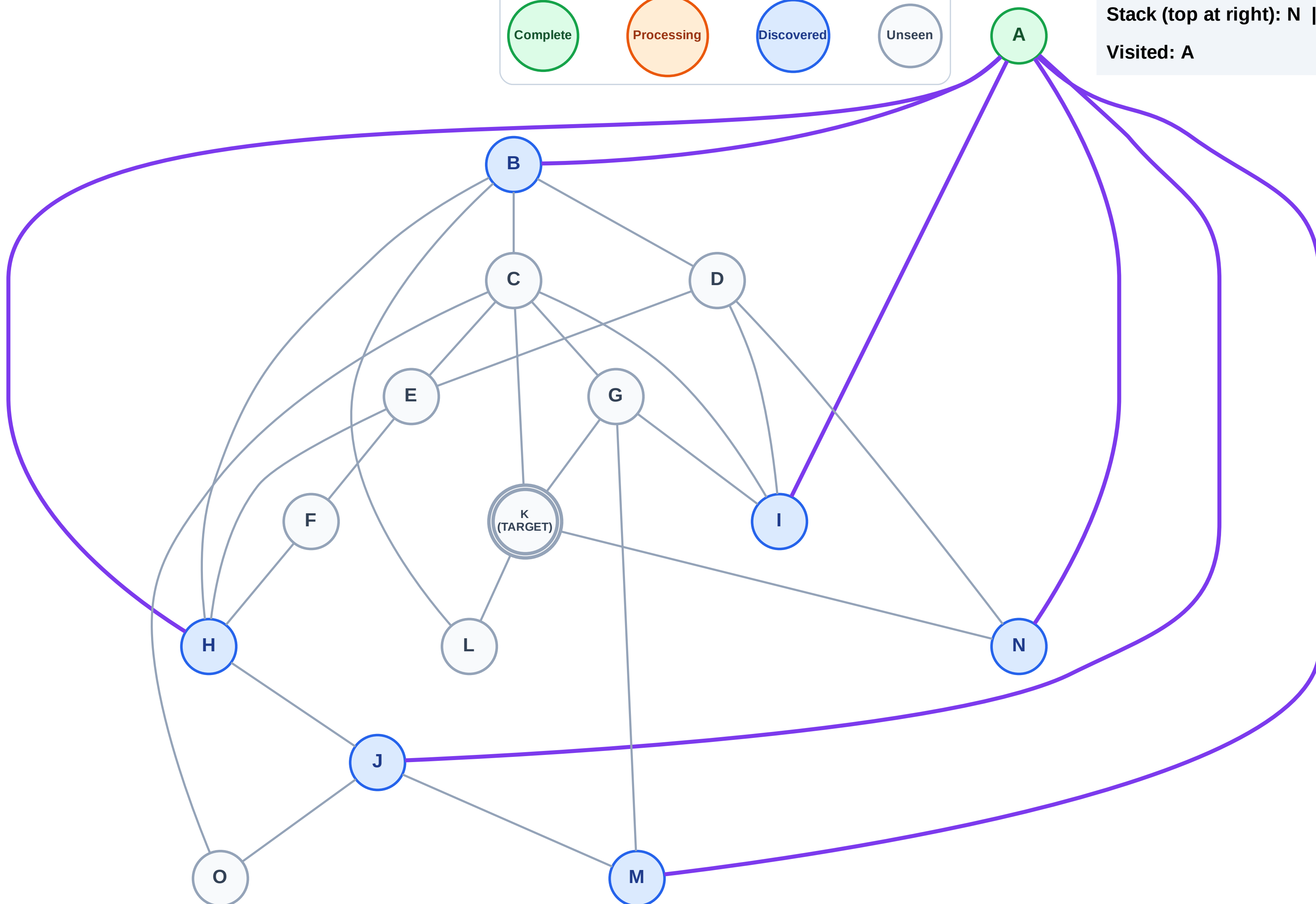
Step 3: Discover N, M, J, I, H, B from A

Legend



Stack (top at right): N | M | J | I | H | B

Visited: A



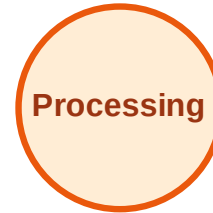
DFS Traversal

Step 4: Process node B

Legend



Complete



Processing



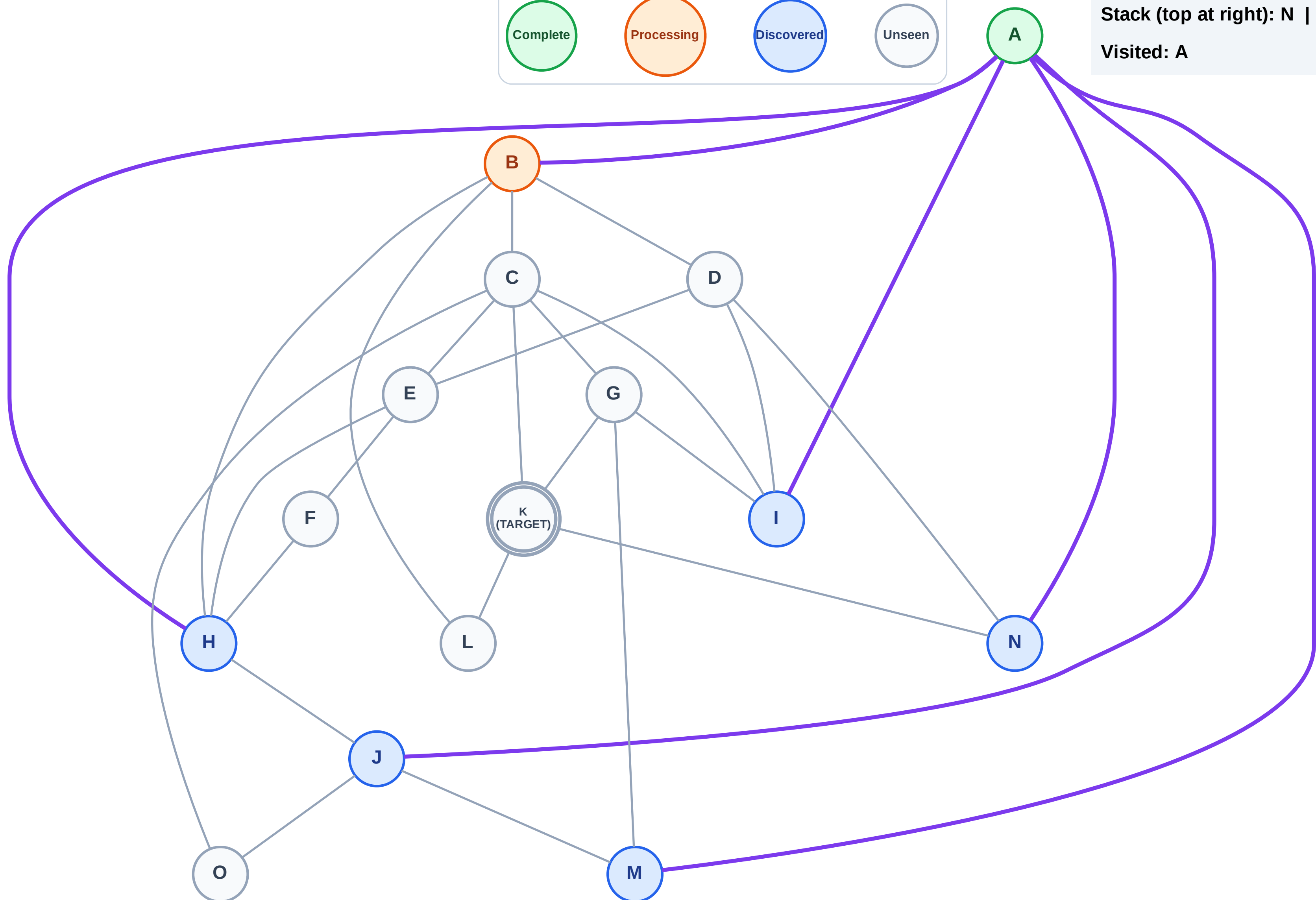
Discovered



Unseen

Stack (top at right): N | M | J | I | H

Visited: A



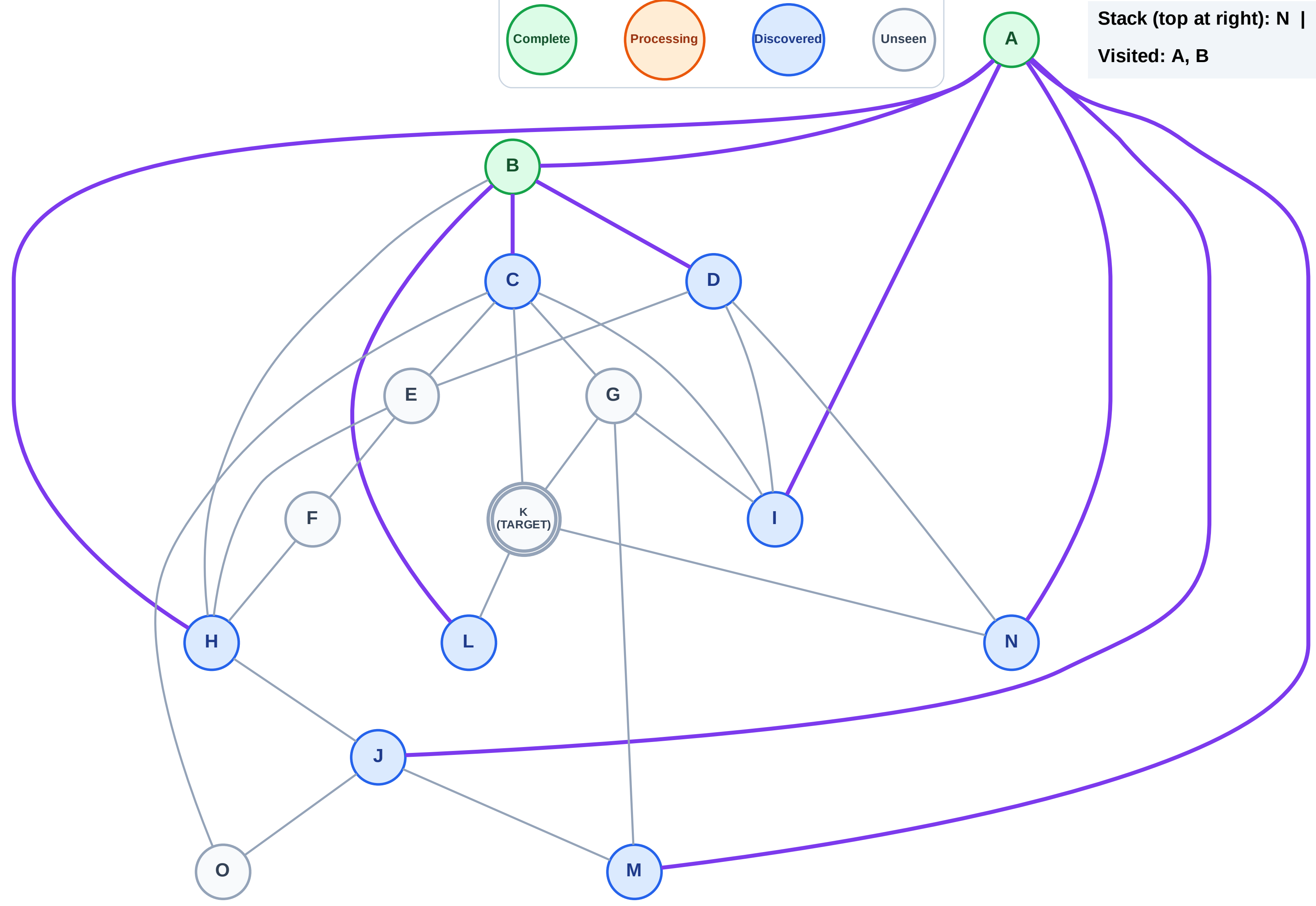
DFS Traversal

Step 5: Discover L, D, C from B

Legend



Stack (top at right): N | M | J | I | H | L | D | C
Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

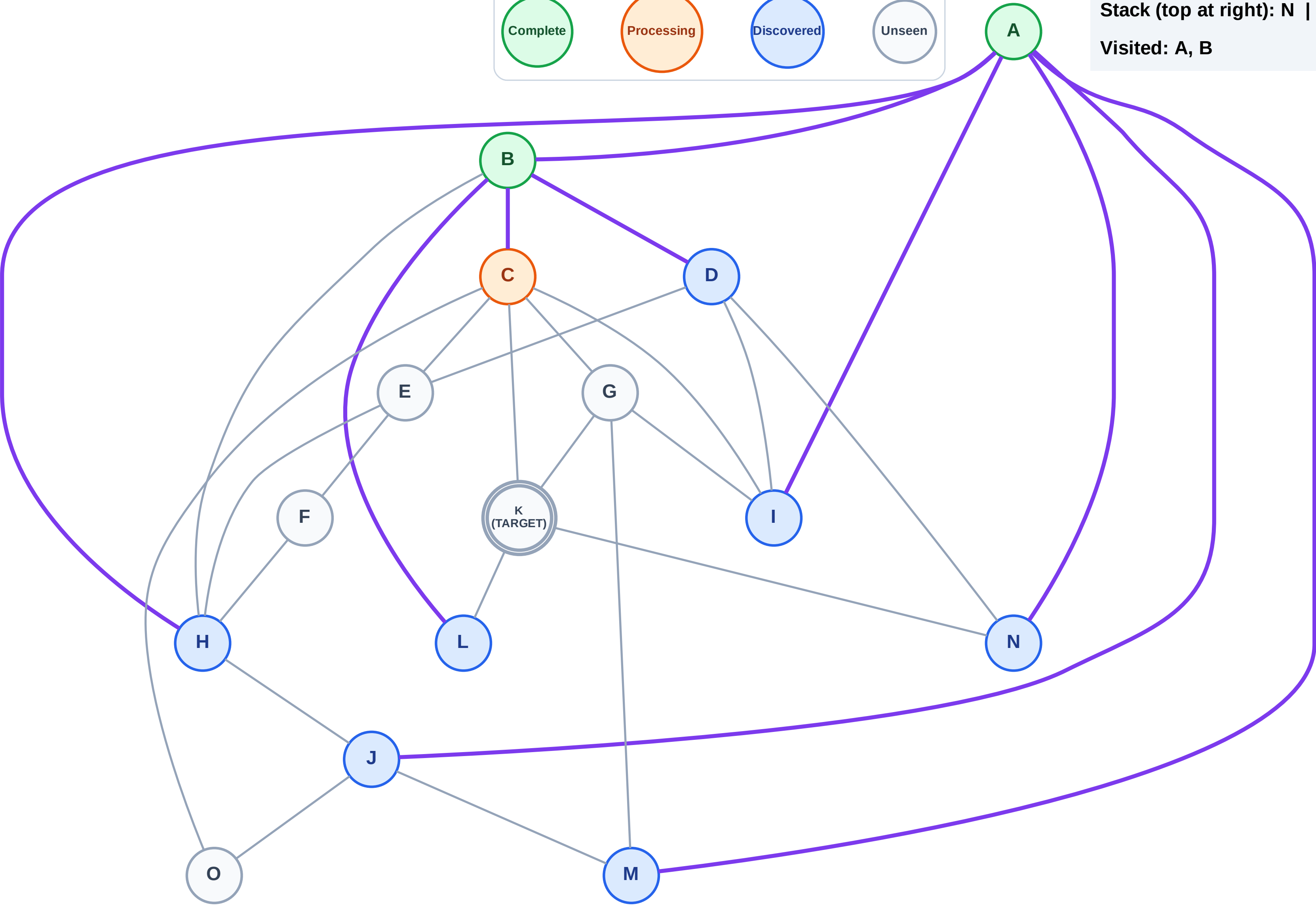
Processing

Discovered

Unseen

Stack (top at right): N | M | J | I | H | L | D

Visited: A, B



Step 7: Discover O, K, G, E from C

Step 7: Discover O, K, G, E from C



DFS Traversal

Step 8: Process node E

Legend

Complete

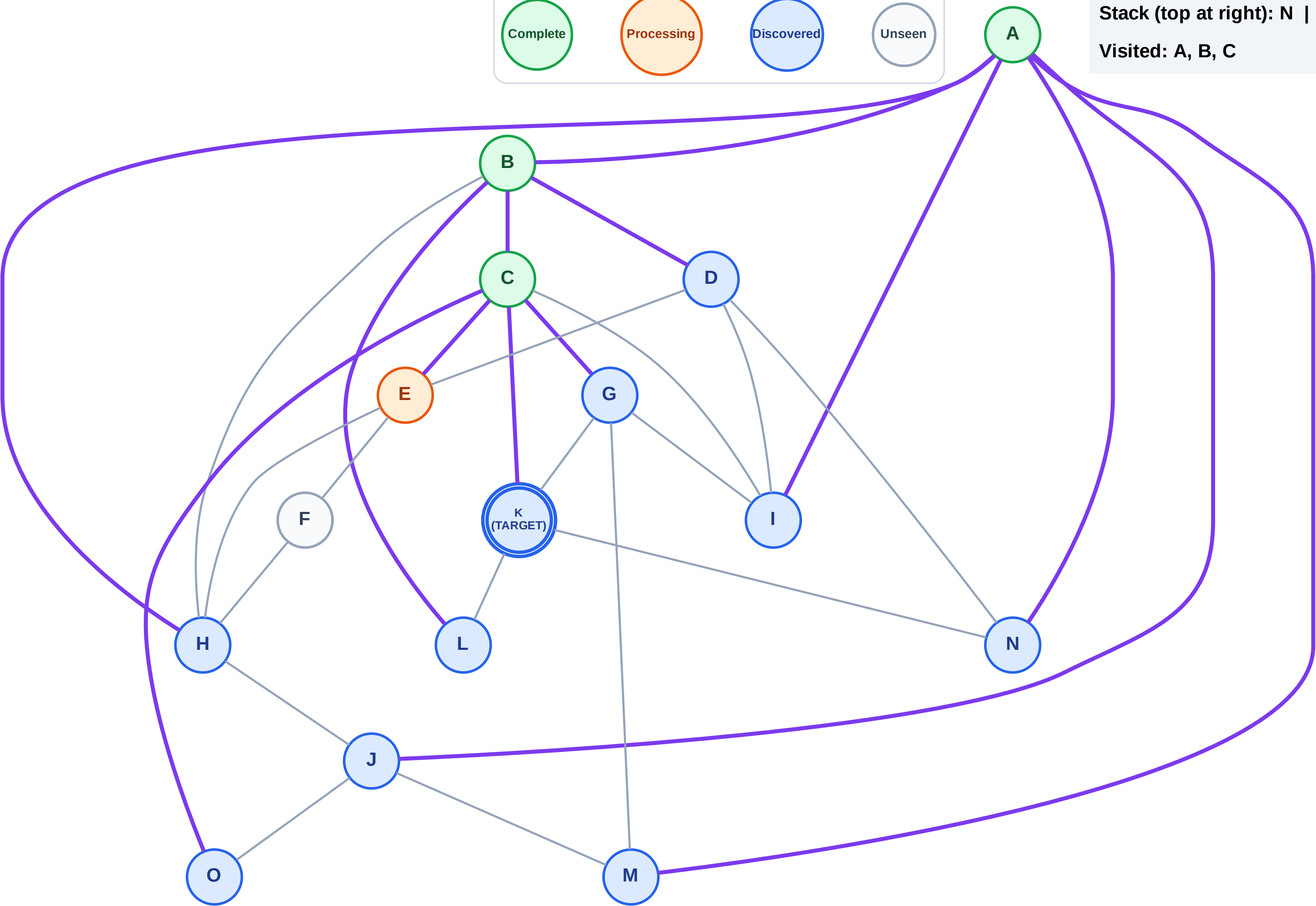
Processing

Discovered

Unseen

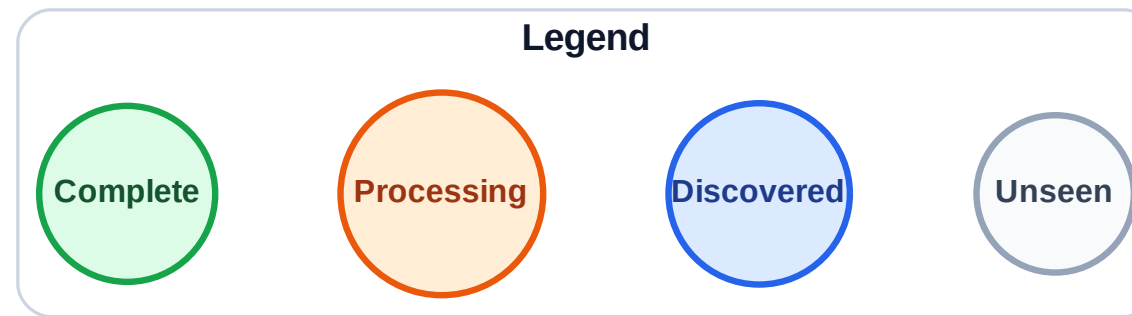
Stack (top at right): N | M | J | I | H | L | D | O | K | G

Visited: A, B, C



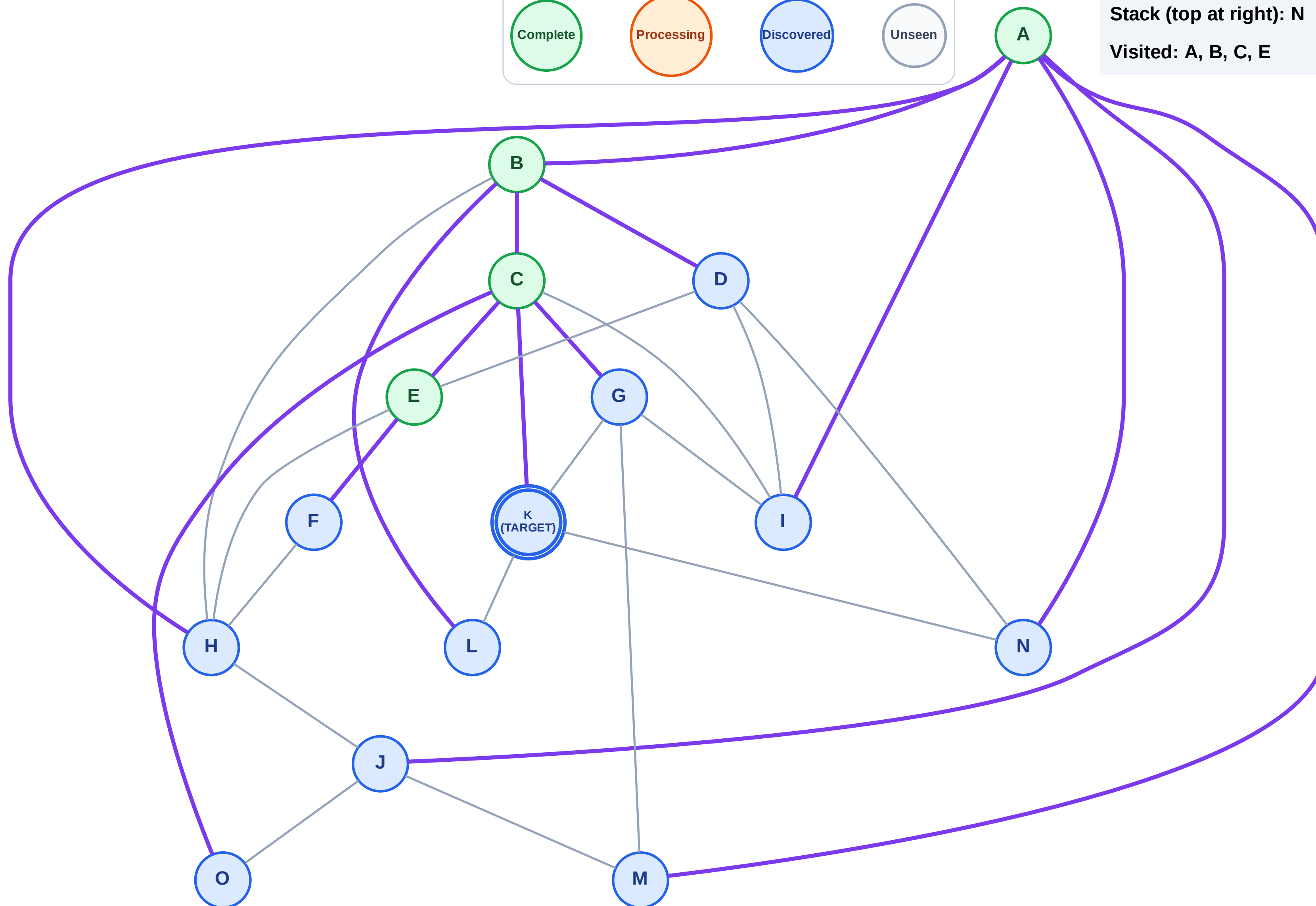
DFS Traversal

Step 9: Discover F from E



Stack (top at right): N | M | J | I | H | L | D | O | K | G | F

Visited: A, B, C, E



DFS Traversal

Step 10: Process node F

Legend

Complete

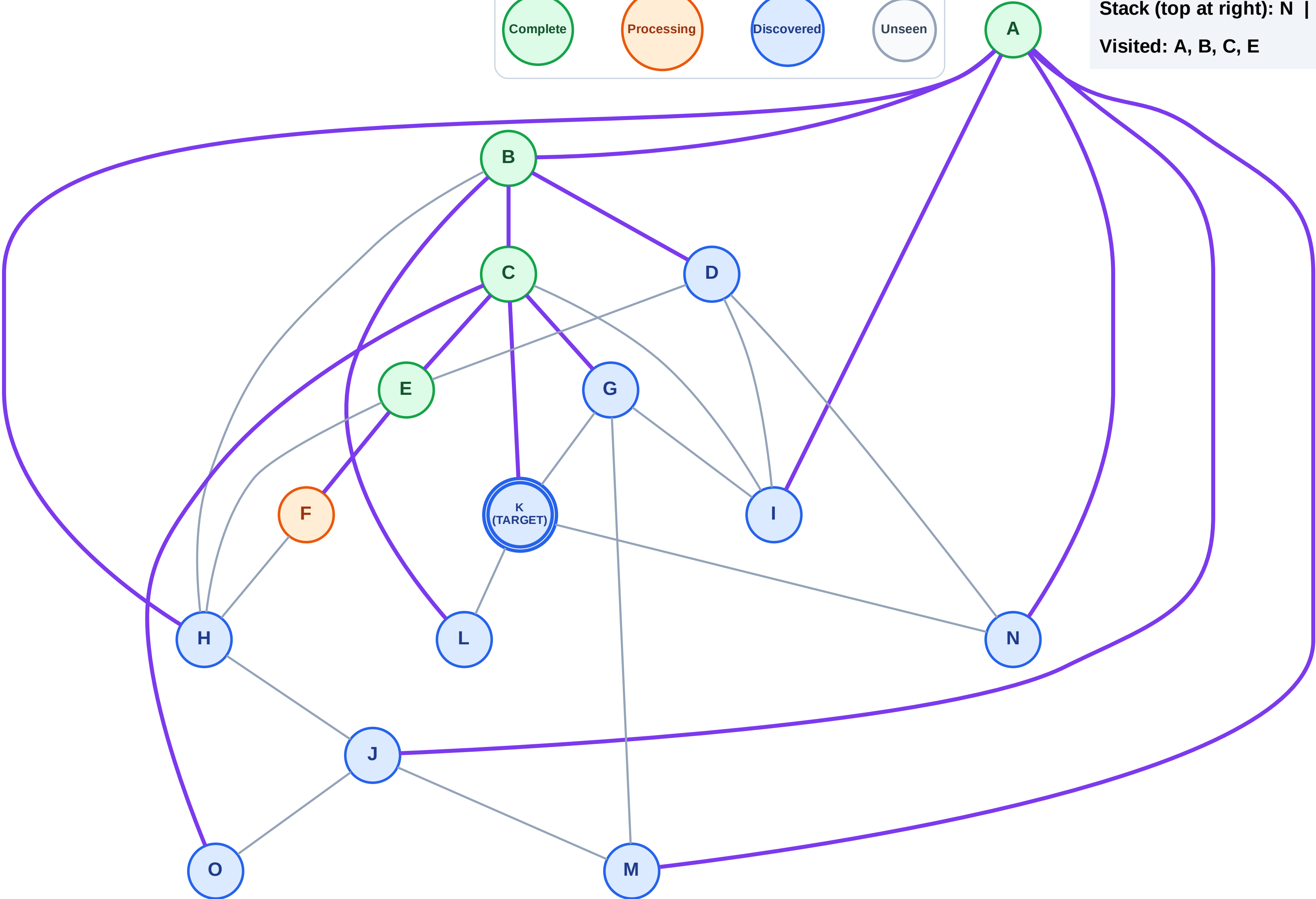
Processing

Discovered

Unseen

Stack (top at right): N | M | J | I | H | L | D | O | K | G

Visited: A, B, C, E



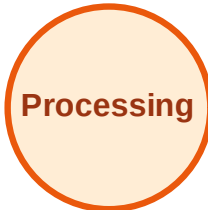
DFS Traversal

Step 11: No new neighbors from F

Legend



Complete



Processing



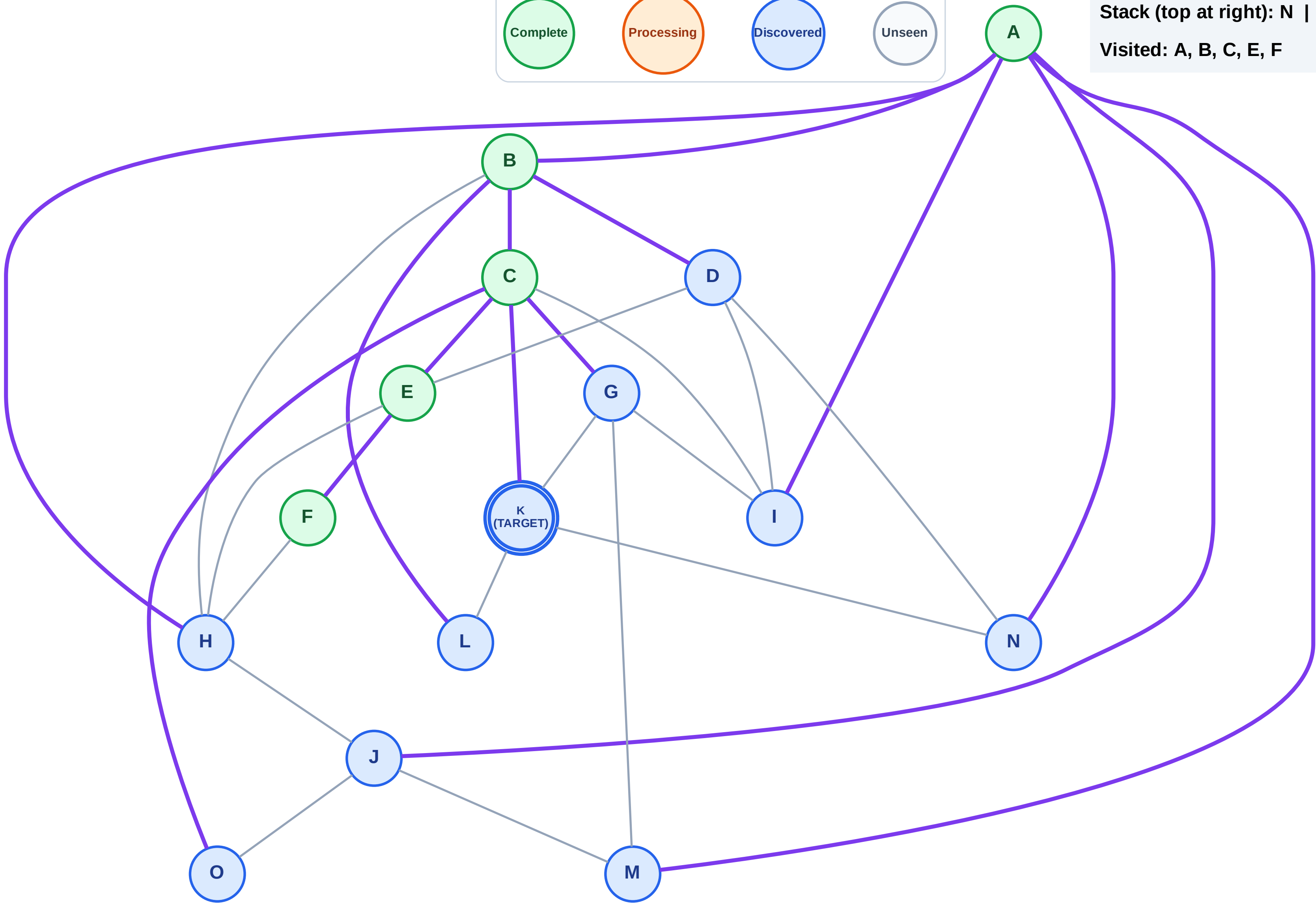
Discovered



Unseen

Stack (top at right): N | M | J | I | H | L | D | O | K | G

Visited: A, B, C, E, F



DFS Traversal

Step 12: Process node G

Legend

Complete

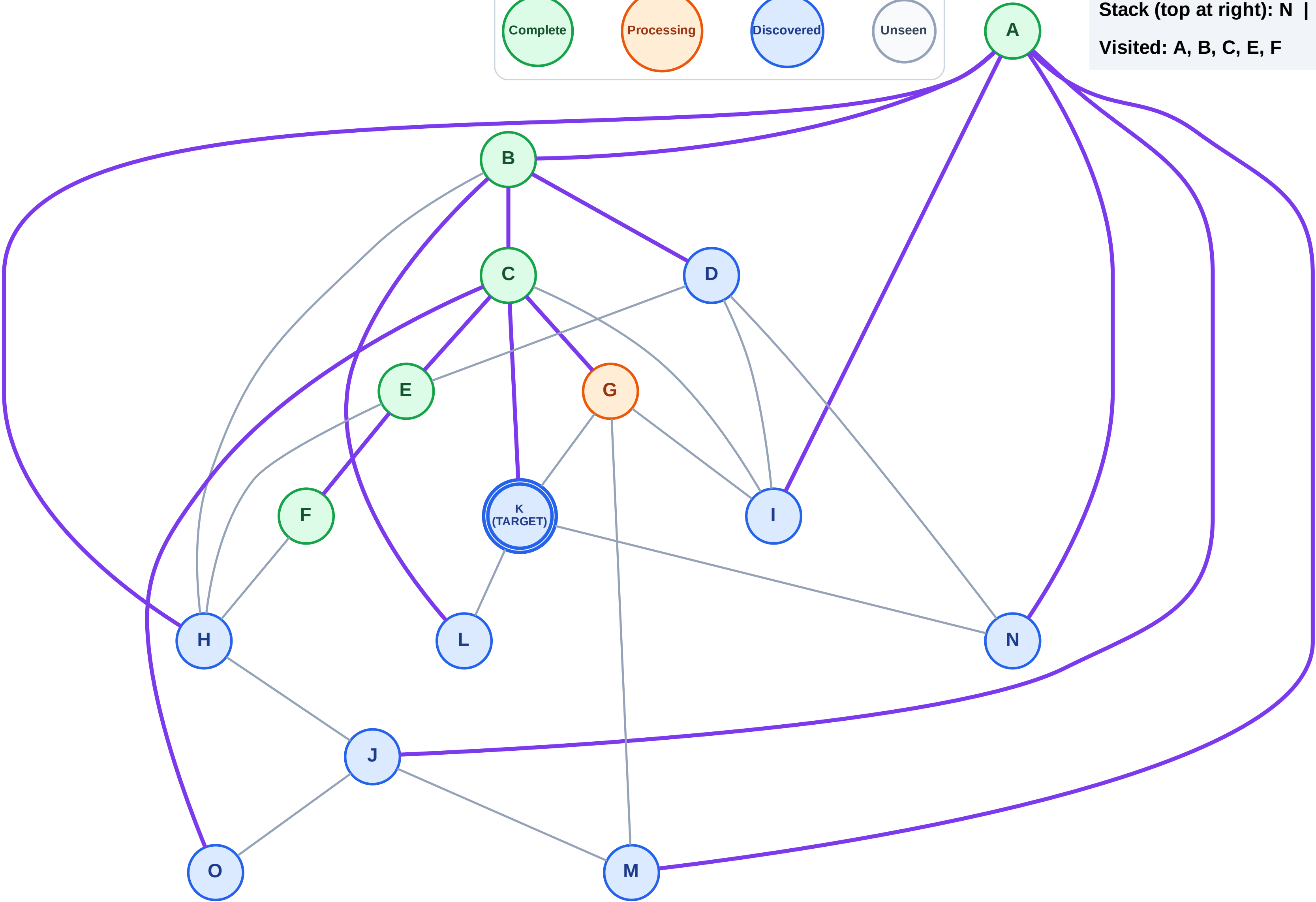
Processing

Discovered

Unseen

Stack (top at right): N | M | J | I | H | L | D | O | K

Visited: A, B, C, E, F



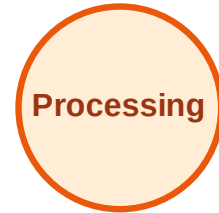
DFS Traversal

Step 13: No new neighbors from G

Legend



Complete



Processing



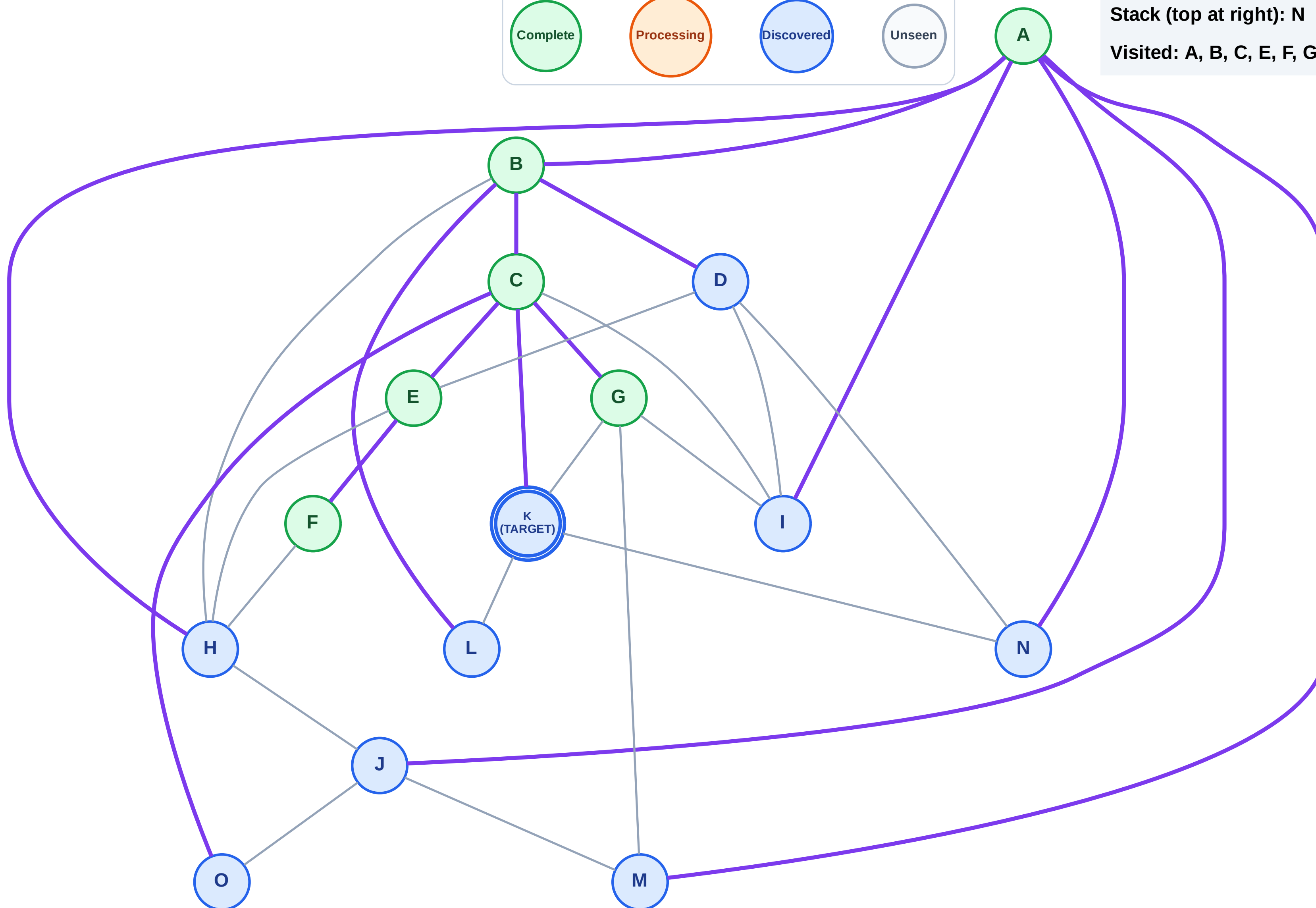
Discovered



Unseen

Stack (top at right): N | M | J | I | H | L | D | O | K

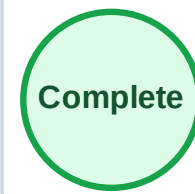
Visited: A, B, C, E, F, G



DFS Traversal

Step 14: Process node K

Legend



Complete



Processing



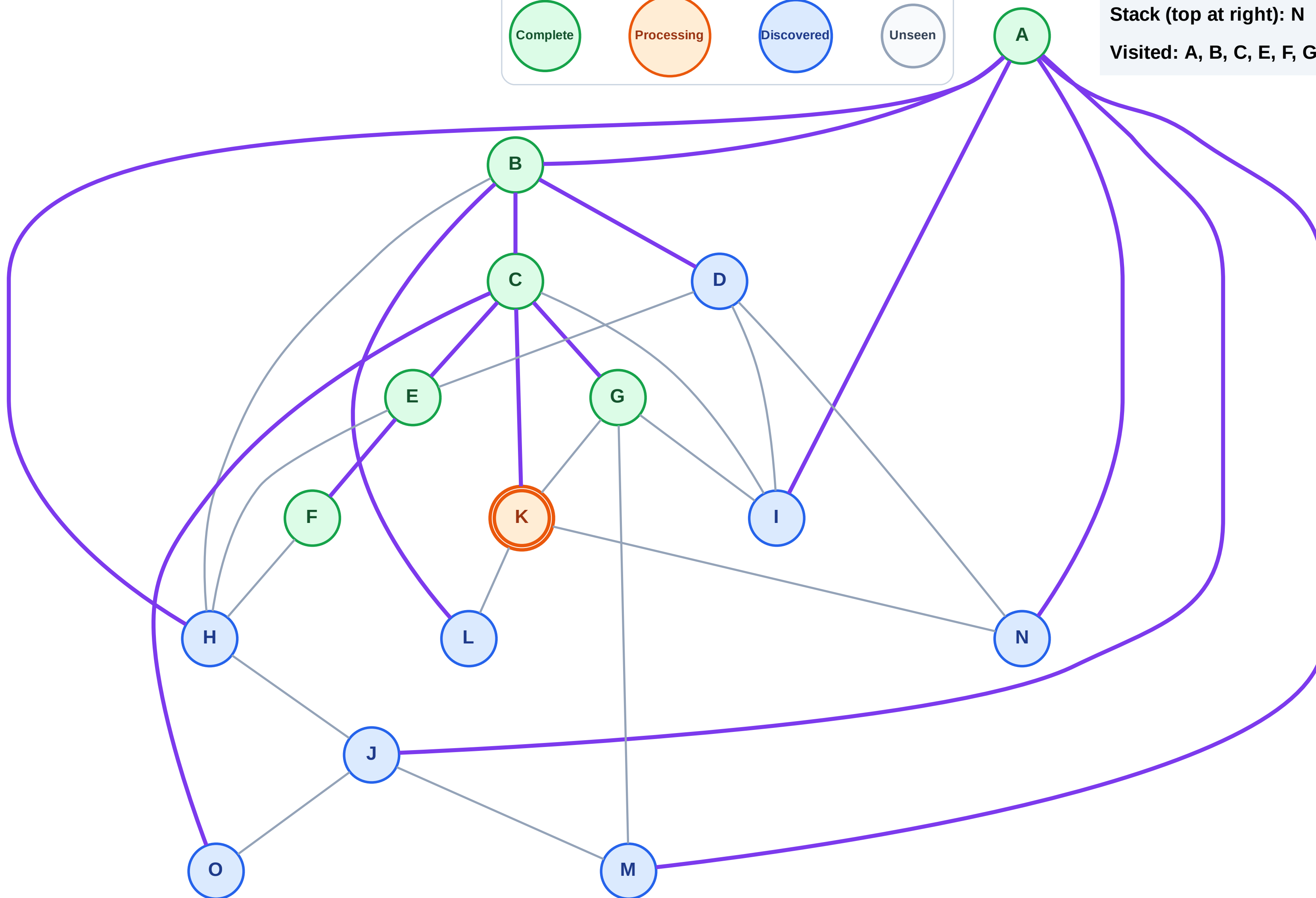
Discovered



Unseen

Stack (top at right): N | M | J | I | H | L | D | O

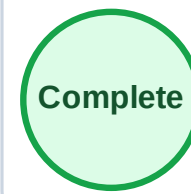
Visited: A, B, C, E, F, G



DFS Traversal

Step 15: TARGET NODE K REACHED! Path (3 hops): A → B → C → K

Legend



Complete



Processing



Discovered



Unseen

Stack (top at right): N | M | J | I | H | L | D | O

Visited: A, B, C, E, F, G, K

